

System Identification and Regression (SIR): Task-conditioned relational discovery in observational systems

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Abstract

Equation discovery usually fixes the observational variables, coordinate system and model class before inference. This hidden representational choice can be more consequential than the optimizer: different coordinates expose different distinctions, and one record can support several exact but scientifically inequivalent relations. We introduce System Identification and Regression (SIR), a task-conditioned framework that treats observations as scientific objects and searches jointly over admissible coordinates and relational forms, with provenance, availability and validation attached to every result. Controlled studies recover classical dynamics, infer a shared law with realization-dependent coefficients, expose non-identifiability from one field realization and show how trajectory-relational lifting separates branches collapsed by projection. In 62 multimodal DIII-D discharges, SIR finds a compact hybrid implicit closure linking energetic, geometric and stability quantities across three signal reconstructions. Algebraic analysis distinguishes legitimate target-containing constraints from tautology while limiting predictive claims. SIR thereby broadens equation recovery into qualified relational discovery.

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Foundation models and tool-using agents are rapidly expanding the scope of automated discovery. Contemporary systems can propose hypotheses, generate and modify executable programs, induce reusable concepts, run computational experiments, refine candidate solutions through automated feedback, and automate portions of the validation and reproducibility auditing of data-analysis workflows [? ? ? ? ?]. These achievements demonstrate that machine-assisted discovery need not be restricted to choosing among a fixed list of human-written models. Yet their apparent open-endedness usually remains conditional on

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a representational substrate supplied in advance: the data have been typed, variables named, executable interfaces defined, admissible operations supplied or induced within a prescribed grammar, and success encoded by an evaluator. The search may be creative within this environment, but the mathematical language in which scientific hypotheses become expressible is still largely inherited.

This limitation becomes especially important when an agent enters a new data universe. A data dictionary states what quantities have been recorded. A scientific ontology states what those quantities mean and how the associated entities are semantically organized. Neither, by itself, determines whether a record should be treated as a trajectory, field, ensemble, event process or coupled collection; which measured or derived quantities should serve as coordinates; which transformations preserve relevant information; which explicit, implicit, differential, integral, stochastic or constraint-based relations should be considered; or what evidence would distinguish a useful relation from an artifact of its construction. These choices constitute a *relational hypothesis language*. They determine not only where a search looks, but what kinds of regularity it is capable of expressing.

Computational discovery has become extraordinarily powerful but methodologically fragmented. SIR attempts to restore a unified discovery cycle in which observations, representations, scientific objectives, and validation are treated as parts of the same computational problem.

SIR is less a competitor to SINDy/PINNs/symbolic regression than a framework in which those become particular discovery strategies.

The same representational bottleneck precedes contemporary agentic systems. Sparse system identification, symbolic regression, Koopman analysis and scientific machine learning recover increasingly complex and interpretable structure from data [1–9]. These methods are most decisive when the state variables, their mathematical roles and the relevant family of equations are sufficiently well specified. Those choices are often treated as preprocessing or problem formulation, although they are themselves scientific hypotheses: they decide which distinctions are visible to the search and which explanations can be represented.

Many contemporary scientific records do not supply a privileged representation. Experimental plasmas, biological measurements, climate observations and industrial sensor systems are heterogeneous projections of processes whose latent variables, observation operators and reconstruction pipelines are only partly accessible. A stored array may admit interpretation as a trajectory, field or ensemble. The same measured quantity may serve as a response in one task, a coordinate in another and part of a constraint in a third. Derived quantities can reveal invariants or separate projected branches, but can also introduce numerical singularities, target dependence or information unavailable at deployment. Several mathematically distinct relations may therefore agree exactly on the same realized object.

The ideal gas law provides a familiar illustration: $[PV=nRT=0.]$ The relation does not intrinsically privilege pressure, volume, temperature or amount of substance. Solving for any one of them produces a useful explicit map, but the underlying scientific statement is relational. The same distinction is consequential in differential-algebraic systems, equations of state, conservation laws and constrained dynamics, where explicit input–output forms may exist only after a coordinate choice or elimination [10–12]. With observational data, however, that choice additionally carries provenance, uncertainty and information-availability conditions. A centered smoother may be admissible for retrospective reconstruction and inadmissible for causal forecasting. A coordinate containing the contemporaneous response may define a legitimate implicit constraint while providing no independent predictive information. A small residual can consequently establish compatibility with a constructed representation without identifying the generator of the record.

Here we introduce System Identification and Regression (SIR), a task-conditioned framework that makes this representational layer explicit and searchable. SIR begins from a provenance-bearing scientific object and couples four decisions that are ordinarily separated: interpretation of the observational object, construction of admissible coordinates, selection of relational forms and validation of the resulting claims. Rather than ending with a static semantic ontology, it associates the observed system with an expandable but contract-bounded relational hypothesis language. Coordinates may be measured, transformed, learned or constructed from the geometry and evolution of the observational object; relations may be explicit, implicit, rational, differential, integral, stochastic or constraint-based. Each candidate remains subject to mathematical well-posedness,

information availability, provenance, dimensional and scientific constraints, and task-appropriate validation.

SIR therefore shifts automated discovery from searching for relations within one prescribed representation to searching over qualified representations themselves. It does not replace symbolic regression, system-identification algorithms, foundation models or scientific agents. Rather, it provides an auditable computational layer in which human- or machine-generated coordinate systems and relational families can be proposed, tested, compared and rejected. Its output is not an unqualified equation, but a relation bound to the coordinates, domain, provenance, evidence and intended use that support it, together with unresolved alternatives when the observations do not justify a unique conclusion.

We demonstrate this distinction through a sequence of controlled and experimental studies. The framework recovers a known dynamical system when the generator is identifiable, discovers a shared relation with realization-dependent coefficients across a heterogeneous population, recognizes when one exact field realization cannot select its generating partial differential equation, and shows how trajectory-relational lifting can separate branches collapsed by projection. We then apply SIR to multimodal DIII-D tokamak observations, where it identifies a compact hybrid implicit closure linking energetic, geometric and stability quantities across multiple signal reconstructions. The resulting account is deliberately more qualified than universal equation recovery: it identifies what mathematical organization a finite observational system supports, what remains non-identifiable, and which additional observations or validation contracts would discriminate among the remaining representations.

For later: There is no task-independent best representation of an observational system. SIR turns that fact into a computational discovery principle: coordinate systems, relations, conditioning requirements and validation criteria are selected jointly under an explicit scientific contract.

1 Results

The Results develop SIR from a general discovery architecture to controlled and experimental instantiations. We first define how a scientific object and discovery contract induce an admissible relational ontology, an explored frontier, and a qualified output. We then show how alternative coordinates can expose or simplify different structures of the same observational record. Controlled studies demonstrate that discovery can terminate in a shared relational family, a task-dependent representation, or a qualified ambiguity class. We next show that conventional model identification is recovered as a restricted SIR problem, whereas allowing representation itself to participate in discovery enlarges the scientific search without changing the downstream estimator. Finally, two distinct contracts applied to multimodal DIII-D observations demonstrate these principles in a heterogeneous experimental system.

1.1 From Scientific Objects to Qualified Representations

Scientific discovery does not begin from a context-free numerical array. As a consequence we represent the *finite scientific object* $\mathcal{O}$ supplied to discovery as:

$$\mathcal{O} = (\mathcal{D}, \Omega_{\text{obs}}, \mathcal{S}, \mathcal{E}, \Pi, \mathcal{A}).$$

Here $\mathcal{D}$ contains the *recorded data objects*, Ω_{obs} their *observed domains*, $\mathcal{S}$ the *sampling*, discretization, and ordering *structure*, $\mathcal{E}$ *uncertainty, missingness, and known error*, Π *provenance and transformation history*, and $\mathcal{A}$ the *conditions under which each component is available*. The data may comprise, for example, trajectories, fields, profiles, event processes, graphs, ensembles, images, meshes, distributions, or coupled and multimodal collections. Retaining these distinctions prevents numerical values from being separated silently from the conditions that determine what can be inferred from them.

A *scientific question* q then imposes a *discovery contract* $\mathcal{K}_q$:

$$\mathcal{K}_q = (q, \mathcal{I}_q, \mathcal{P}_q, \mathcal{B}_q, \mathcal{H}_q, U_q, \mathcal{V}_q, \Omega_q).$$

The *information operator* $\mathcal{I}_q$ selects the portion of the scientific object that may be used for the task. The *admissibility rules* $\mathcal{P}_q$ encode semantic, dimensional, mathematical, epistemic, and operational requirements; the *contract boundary* $\mathcal{B}_q$ bounds the complexity, depth and computational extent of permitted constructions and search; $\mathcal{H}_q$ specifies admissible external scientific knowledge that may inform hypothesis proposal or search priority; $\mathcal{U}_q$ specifies *what constitutes a useful scientific result*; $\mathcal{V}_q$ specifies the *validation contract* required to support that result; and Ω_q is its *intended domain*. The external knowledge resource $\mathcal{H}_q$ may be empty or may contain, for example, peer-reviewed literature, curated databases, established models, laboratory log-books, domain knowledge, or previously qualified discovery results. Its sources, versions, retrieval procedures and transformations form part of discovery provenance. Retrospective reconstruction, causal prediction, compression, mechanism identification, and closed-loop control can therefore impose different contracts on the same stored record.

The information boundary first induces a *task-admissible record* $\mathcal{O}_q = \mathcal{I}_q(\mathcal{O})$, which is then assigned a *task-conditioned mathematical interpretation* $X_q = \text{Obj}_q(\mathcal{O}_q)$. Here X_q specifies the mathematical object through which the admissible record is to be interrogated, while the scientific objective itself is specified by the discovery contract $\mathcal{K}_q$. This assignment is consequential. For example, the same stored array may be interpreted as a field, a trajectory through a function space, an ensemble of states, or another structured object, and these interpretations admit different operators, coordinate constructions, and scientific claims. Object interpretation therefore precedes coordinate construction rather than being treated as a neutral preprocessing decision.

The contract does not act directly on an unrestricted feature universe. SIR first constructs a task-conditioned *relational ontology* $\mathcal{G}_q$, such that

$$\mathcal{G}_q = \Gamma(X_q, \mathcal{O}_q, \mathcal{K}_q; \Lambda, \mathfrak{T}),$$

where Λ is an extensible catalogue of typed coordinate and computational constructors and $\mathfrak{T}$ an extensible catalogue of relation and model templates. Constructors may include elementary mathematical operations such as differentiation, integration, projection, and nondimensionalization, but may also wrap learned maps, numerical solvers, scientific codes, simulators, inverse procedures or other callable computational models whose outputs can participate in subsequent coordinate or relational construction. These catalogues provide a vocabulary rather than a search space and are not assumed to be exhaustive. The ontology $\mathcal{G}_q$ provides the task-conditioned grammar: it contains constructions whose type signatures and declared preconditions make them candidates for admissibility under the supplied object and task. Data-dependent conditions such as numerical stability, denominator support, uncertainty amplification, or operational availability are subsequently evaluated using evidence permitted by the discovery contract.

Admissibility does not imply that all reachable constructions should receive equal search effort. SIR therefore distinguishes the relational ontology from the policy used to explore it. We denote a task-conditioned search policy by $\Sigma_q = \text{SearchPolicy}(\mathcal{G}_q, \mathcal{K}_q)$. One component of such a policy may be a proposal priority $\pi_q(\xi \mid X_q, \mathcal{G}_q, \mathcal{H}_q)$ associated with an admissible construction ξ . Scientific precedent in $\mathcal{H}_q$ can thereby favor coordinate families, operator compositions, or relational templates that have proved useful in analogous systems, while computational cost, conditioning, redundancy, uncertainty and previously observed development-stage utility can modify their priority. The quantity π_q is schematic: it may represent a score, ordering, proposal distribution, learned policy, or other mechanism for allocating search effort. It should be noted that search precedent is not validation evidence. The Σ_q governs where the admissible space is explored first, whereas $\mathcal{U}_q$ and $\mathcal{V}_q$ determine how the resulting candidates are scientifically compared and qualified. Low-priority admissible regions may remain accessible so that prior knowledge accelerates discovery without excluding genuinely novel representations.

Concretely, each coordinate c carries an attributed type τ ;

$$\tau(c) = (\text{kind}(c), \text{dim}(c), \text{unit}(c), \Omega_c, \mathcal{S}_c, \mathcal{E}_c, \mathcal{A}_c, \Pi_c, \dots),$$

recording its semantic role, physical dimensions, units, supported domain, sampling or regularity, uncertainty, information availability and provenance. The type is deliberately extensible: additional attributes may encode, for example, tensor character, coordinate frame, symmetry, topology, fidelity level, or other object-specific structure when required. A relational constructor $\lambda \in \Lambda$ has an input-output type signature and an

admissibility predicate,

$$\lambda : (\tau_1, \dots, \tau_r) \longrightarrow \tau_{\text{out}}, \quad P_\lambda(X_q, \mathcal{O}_q, \mathcal{K}_q; c_1, \dots, c_r) = 1.$$

Composition is permitted only when these conditions are satisfied. A quotient, for example, requires adequate denominator support and acceptable propagated uncertainty; a logarithm requires an admissible domain; an integral requires compatible support and boundary conventions; and a centered temporal operator may be admissible for retrospective reconstruction while being inadmissible for causal prediction.

Importantly, a scientific coordinate and its numerical realization are distinct. The operator $\text{Rate}(y)$, for example, specifies a temporal-rate coordinate, not a particular finite-difference formula. For an effectively noise-free simulated trajectory, direct differentiation may be admissible. For a noisy experimental record, the same semantic coordinate may instead require local polynomial estimation, filtering, smoothing, or regularized differentiation, or may be rejected if no realization satisfies the uncertainty requirements of the contract. Thus the observational error model can alter not merely which candidate wins, but which constructions are scientifically admissible to begin with. The same distinction applies more generally to derivatives, integrals, quotient coordinates, projections, latent embeddings, model-derived quantities and other constructed objects: the scientific meaning of a coordinate is separated from the numerical procedure used to realize it.

Starting from the source coordinates induced by the task-admissible object, $\mathcal{N}_q^{(0)} = \text{Src}(X_q)$, one realization of the ontology construction can expand the reachable coordinate set recursively,

$$\mathcal{N}_q^{(d+1)} = \mathcal{N}_q^{(d)} \cup \left\{ \lambda(c_1, \dots, c_r) : \begin{array}{l} c_i \in \mathcal{N}_q^{(d)}, \\ \lambda \in \Lambda, \\ P_\lambda(X_q, \mathcal{O}_q, \mathcal{K}_q; c_1, \dots, c_r) = 1, \\ \text{cost}_q[\lambda(c_1, \dots, c_r)] \leq \mathcal{B}_q \end{array} \right\}.$$

Here cost_q may encode operator depth, derivative order, interaction order, numerical burden, or other contract-specific complexity controls. Provenance edges retain how every derived quantity was constructed. The recursion is illustrative rather than restrictive: other grammar expansion, program synthesis, agentic, symbolic, probabilistic, or learned procedures may instantiate the same ontology construction provided that admissibility and provenance remain explicit.

The resulting ontology induces the concrete admissible coordinate and relational spaces,

$$\mathfrak{C}_q = \text{Coord}(\mathcal{G}_q), \quad \mathfrak{R}_q(C) = \text{Rel}(\mathcal{G}_q, C),$$

and hence the admissible coordinate–relation universe

$$\mathfrak{A}_q = \{(C, R) : C \in \mathfrak{C}_q, R \in \mathfrak{R}_q(C)\}.$$

Ontology construction is therefore distinct from model selection: semantic compatibility, observability, uncertainty, information availability, and the declared task determine what may be searched before target performance is used to rank qualified candidates. Figure 1 illustrates representative coordinate constructors and relational families that may populate such an ontology-induced admissible universe; the categories are intentionally overlapping and non-exhaustive.

The admissible universe $\mathfrak{A}_q$ may nevertheless be combinatorially large or effectively unbounded, and SIR does not require exhaustive enumeration. The search policy instead generates an auditable explored frontier,

$$\widehat{\mathfrak{A}}_q = \text{Explore}(\mathfrak{A}_q; \Sigma_q, \mathcal{B}_q) \subseteq \mathfrak{A}_q.$$

This distinction separates scientific admissibility from computational exploration. A construction absent from $\widehat{\mathfrak{A}}_q$ has not thereby been shown inadmissible, inferior or nonexistent; it may simply not have been reached under the instantiated grammar, search policy, and computational bounds. Negative discovery claims are consequently conditional on the admissible language and explored frontier that produced them.

Additionally, the ontology need not remain static during development. Candidate constructions may expose singularities, redundancy, instability or poor conditioning, poor uncertainty propagation or, conversely, stable

composite quantities that are scientifically useful enough to promote to new coordinates. SIR therefore permits provenance-preserving ontology refinement,

$$\mathcal{G}_q^{(0)} \longrightarrow \mathcal{G}_q^{(1)} \longrightarrow \dots \longrightarrow \mathcal{G}_q^{(m)} \equiv \mathcal{G}_q,$$

with corresponding refinement of the admissible coordinate and relation spaces, the admissible universe and, where appropriate, the search policy. Such refinement may use development evidence, domain knowledge, symbolic simplification, or other admissible discovery procedures, but data reserved for final validation must not influence the revised ontology or search policy. When protected evaluation is required, $\mathcal{V}_q$ distinguishes development evidence $\mathcal{O}_q^{\text{dev}}$ from protected evaluation evidence $\mathcal{O}_q^{\text{eval}}$; ontology construction, refinement, and adaptive search may access only the former. If no protected split is required, $\mathcal{O}_q^{\text{dev}}$ may coincide with the complete task-admissible record. Changes to $\mathcal{H}_q$, $\mathcal{G}_q$, Σ_q or the explored frontier remain part of provenance. This permits iterative scientific reasoning without collapsing ontology construction or search into post hoc optimization on the evaluation set.

Conditional on the explored admissible frontier, SIR selects

$$(C^*, R^*) \in \text{Select}_q(\hat{\mathfrak{A}}_q; U_q, \mathcal{V}_q).$$

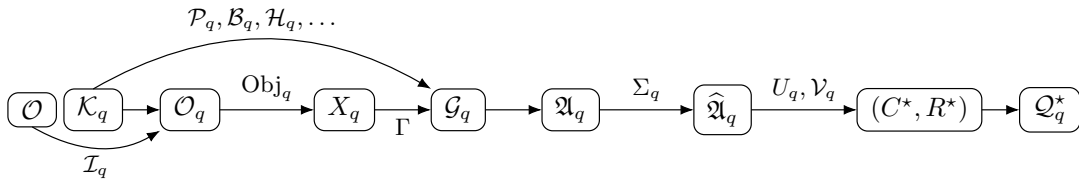
Here U_q evaluates the coordinate–relation representation jointly rather than the relation in isolation; schematically, its arguments include $U_q(C, R; \mathcal{O}_q^{\text{dev}}, \mathcal{V}_q)$. The notation U_q does not require every scientific preference to be collapsed into a single weighted scalar. A discovery contract may instead impose hard admissibility constraints, partial or lexicographic orderings, tolerance-defined equivalence classes, Pareto criteria, or combinations of these, etc. When a scalar utility is appropriate, Select_q may reduce to an ordinary argmax over $\hat{\mathfrak{A}}_q$. The essential requirement is that the rule by which candidates are qualified and compared is declared as part of $\mathcal{K}_q$, rather than chosen after viewing protected evaluation evidence.

The resulting representation is therefore not simply the relation with the smallest residual. It is a scientific hypothesis selected from the portion of relational space that the object, contract, and realized search make scientifically and computationally accessible. Its qualification travels with the result. We write the qualified output schematically as

$$\mathcal{Q}_q^* = (\mathcal{K}_q, C^*, R^*, \Omega^*, \Pi^*, \mathcal{A}^*, \mathcal{E}_{\text{val}}^*, \mathcal{L}^*),$$

where Ω^* denotes the *supported domain*, Π^* the *provenance of the selected representation*, $\mathcal{A}^*$ its *information-availability conditions*, $\mathcal{E}_{\text{val}}^*$ the *validation evidence* supporting the claim, and $\mathcal{L}^*$ its *unresolved limitations and qualifications*. The provenance Π^* includes, directly or by immutable reference, the ontology version, external-knowledge snapshot, numerical realizations, search policy, and explored frontier from which the result arose. Retaining $\mathcal{K}_q$ with the result makes explicit that the same coordinate–relation pair can support different scientific claims under different information, utility, or validation contracts. When the observations do not distinguish several task-equivalent representations, SIR retains the corresponding qualified model set or equivalence class rather than silently promoting one member to a uniquely identified law.

This hierarchy can be summarized schematically as:



This construction makes explicit the organizing principle of SIR: *relational ontology precedes relational search*. SIR does not enumerate an unrestricted universe of expressible formulas and subsequently regularize the result. It constructs a scientifically reachable relational universe from the scientific object and discovery contract, allocates finite search effort within that universe through an explicit and auditable search policy, and qualifies the result according to the evidence required by the task. Object interpretation therefore precedes coordinate construction, ontology precedes search, search priority guides exploration rather than scientific

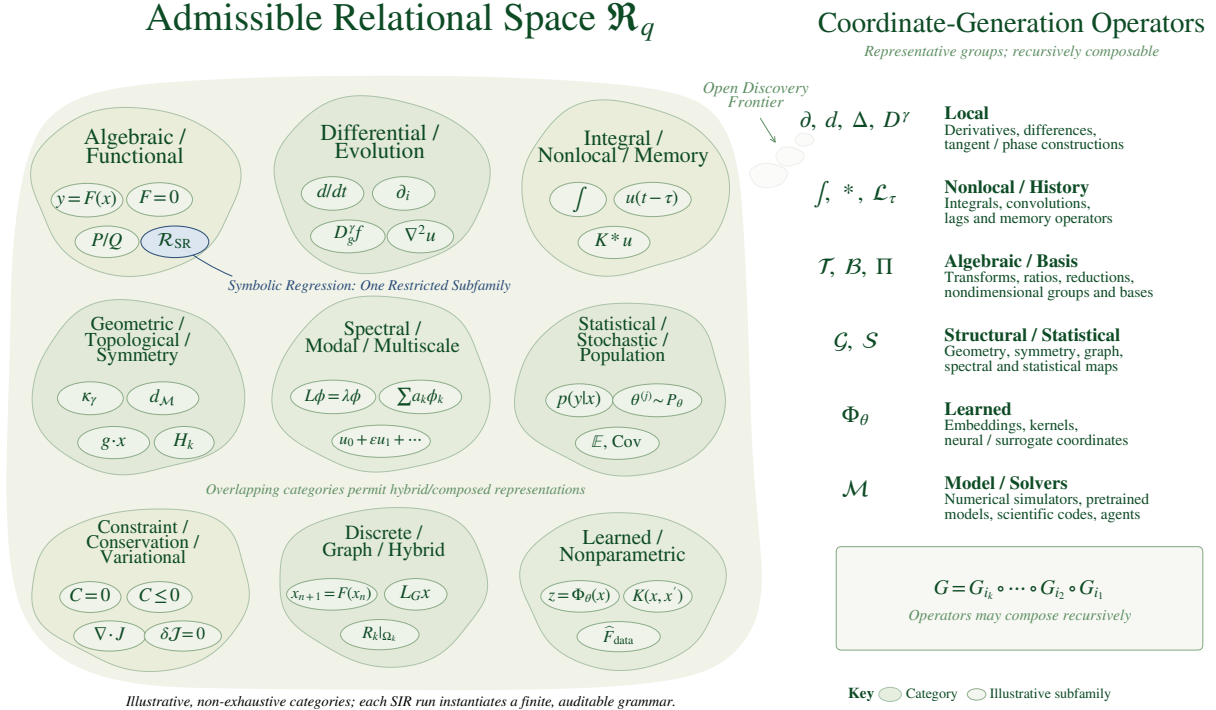


Figure 1: Admissible relational space in SIR. The task-conditioned search space $\mathfrak{R}_q$ spans overlapping families of algebraic, differential, nonlocal, geometric, spectral, statistical, variational, discrete, and learned relations; symbolic regression is one restricted subfamily. Coordinate-generation operators may be combined recursively to construct additional admissible representations. The illustrated categories are representative rather than exhaustive, leaving an open discovery frontier beyond any finite instantiated grammar.

truth, and qualification bounds the scientific claim attached to the selected representation. Different scientific questions, information boundaries, prior knowledge, or observational assumptions may consequently induce different ontologies, explored frontiers, and qualified representations of the same underlying record.

1.2 Object-adapted Coordinates Expand the Searchable Representation

The coordinates recorded by an observational system need not be those in which its scientifically relevant organization is most simply, or conventionally expressed. Once the task-admissible record has been assigned a mathematical interpretation X_q , that object determines which coordinate constructors in the relational ontology $\mathcal{G}_q$ are meaningful/admissible. A trajectory may admit rates, delays, integrals, tangent quantities, recurrence measures, or local geometric constructions; a field may admit gradients, fluxes, modal coefficients, weak-form functionals, or multiscale representations; an event process may admit waiting times, marks, intensities, or hazards; and an ensemble may admit moments, empirical measures, parameter distributions, or realization-level summaries. Algebraic, symmetry-adapted, dimensionless, model-derived, and learned constructions can provide additional coordinates across several object classes. These examples are illustrative rather than exhaustive. In SIR, such constructions are not neutral preprocessing choices: they are typed hypotheses about how the task-admissible object should be represented.

A coordinate constructor can therefore change which distinctions of X_q are exposed to a subsequent relational search. Some constructions compress variation, others separate states that coincide under a lower-dimensional projection, and still others reorganize a scientific target around a simpler geometric, statistical, or dynamical structure. Conversely, a construction can discard distinctions relevant to the task or introduce singularities, unstable inverse maps, excessive uncertainty, or information unavailable at the required operational boundary.

The ontology determines whether such a construction is admissible; the search policy Σ_q determines whether and when it enters the explored frontier $\widehat{\mathfrak{A}}_q$; and its eventual usefulness is judged only through the task-specific comparison and validation rules. Coordinate construction is thus part of representational discovery rather than an unrestricted expansion of a feature library.

Trajectory-relational coordinates provide one concrete example. Let $\gamma : I \rightarrow \mathcal{M}$ be a differentiable trajectory and $f, g : \mathcal{M} \rightarrow \mathbb{R}$ differentiable observables. Wherever $\partial_\gamma g \neq 0$, define

$$D_g^\gamma f = \frac{\partial_\gamma f}{\partial_\gamma g}. \quad (1)$$

We refer to $D_g^\gamma f$ as a first-order *trajectory-relational derivative*, or, for brevity throughout this work, a *phase derivative*. Locally, Eq. (1) is the ordinary derivative of $f \circ \gamma$ with respect to $g \circ \gamma$. It therefore describes how one observable changes relative to another along the same realized trajectory, without requiring a globally single-valued relation $f = F(g)$ and without assigning causal direction. Under a regular reparameterization of the trajectory, the common traversal-rate factor cancels. The phase derivative is consequently a local relational coordinate along the trajectory rather than a temporal derivative or an assertion that f is globally a function of g .

The distinction established in Section 1.1 between a scientific coordinate and its numerical realization is particularly important for this family. The semantic construction $D_g^\gamma f$ is defined on the portion of the trajectory for which the reference derivative is regular; a sampled realization additionally requires a declared derivative estimator, boundary convention, and small-denominator policy. Alternative shifted, bounded, piecewise, or sensitivity-centered constructions may provide useful finite coordinates on other domains, but they are distinct members of the ontology rather than interchangeable numerical implementations of Eq. (1). Their domains, conditioning, transformation properties, and numerical realizations therefore travel with their provenance.

Figure 2 illustrates how different admissible coordinates interrogate different aspects of the same observational object. Two independently prescribed observables, $x_1(t)$ and $x_2(t)$, are unchanged throughout all panels, while the distinction of interest is the local branch of the target x_1 , indexed here by the sign of $\dot{x}_1$. The level-rate representation $(x_2, \dot{x}_2)$ is a familiar phase-plane construction: it describes the instantaneous level and rate of the reference observable, but the increasing and decreasing branches of x_1 remain strongly folded in this projection (Fig. 2b). The sensitivity-centered trajectory-relational representation $(\dot{x}_2, D_{x_2}^{\text{sc}} x_1)$ instead describes the evolution of x_1 relative to that of x_2 , and reorganizes the same samples around the exact target turning manifold associated with $\dot{x}_1 = 0$ (Fig. 2c). The rate-rate representation $(\dot{x}_2, \dot{x}_1)$ discards the instantaneous levels and exposes the joint local evolution of the two observables directly (Fig. 2d), whereas the rate-acceleration representation $(\ddot{x}_1, \dot{x}_1)$ is internal to the target itself, indexing its turning events by their local acceleration and thereby exposing structure in the target’s own differential evolution (Fig. 2e).

These representations are therefore complementary rather than merely alternative visualizations. Each preserves and emphasizes different information: state relative to rate, evolution relative to another observable, co-evolution of rates, or higher-order local structure within one observable. The phase derivative construction has an additional simplifying effect in this example. A complicated folded branch organization in the level-rate coordinates is reorganized as variation about a simple, explicitly characterized turning manifold. In this sense it partially *flattens* the representation: subtracting the turning manifold places the target condition $\dot{x}_1 = 0$ at zero, converting a geometrically complicated distinction into a low-complexity relational one. No new measurement has been introduced; the simplification results from changing the coordinates in which the same task-admissible object is represented.

¹For this illustrative example, the two observables are defined analytically as

$$\begin{aligned} x_1(t) &= 0.95 \sin(0.72t + 0.28 \sin(0.13t)) + 0.34 \sin(2.05t + 0.7) \\ &\quad + 0.18 \sin(4.70t - 0.3) + 0.16 \sin(0.17t) \sin(1.33t + 0.4) + 0.08 \sin(7.40t + 1.1), \\ x_2(t) &= 0.82 \sin(0.49t + 0.45) + 0.31 \sin(1.18t - 0.35) \\ &\quad + 0.14 \sin(2.66t + 0.25) + 0.08 \sin(5.10t - 0.6). \end{aligned}$$

The signals are independently prescribed functions of the common parameter t ; no functional dependence $x_1 = F_1(x_2)$ or $x_2 = F_2(x_1)$ is assumed.

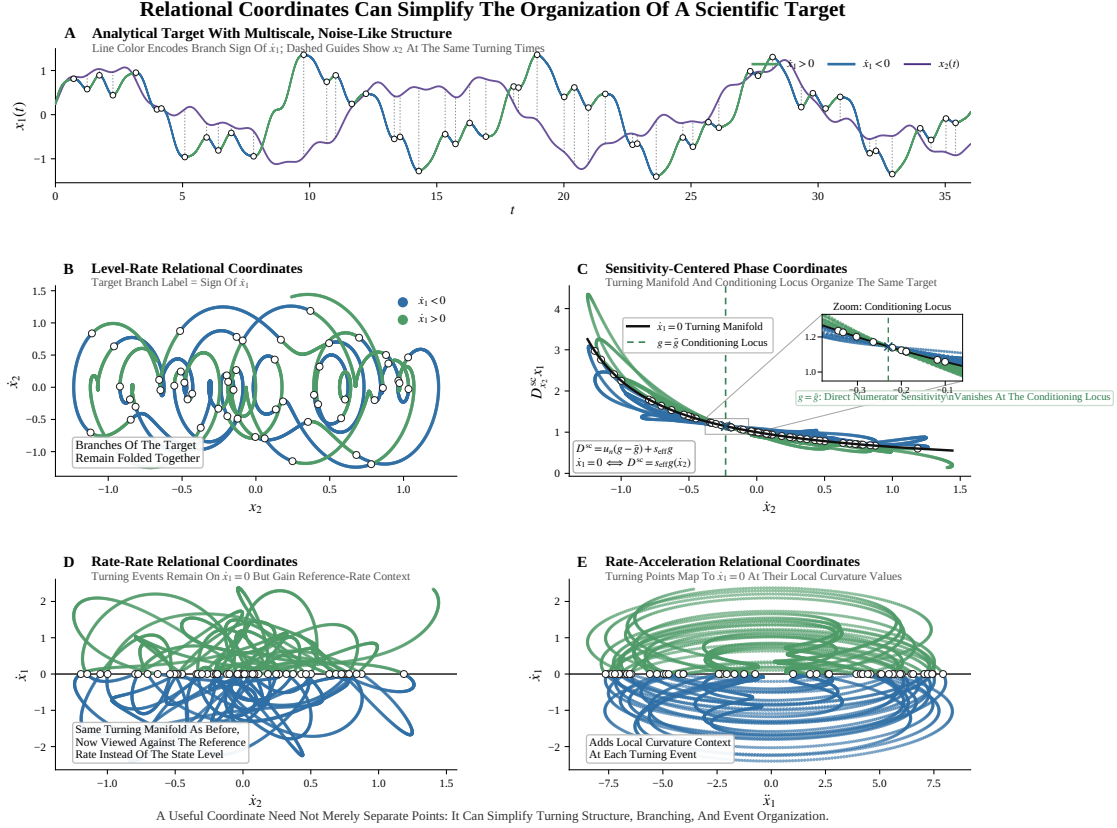


Figure 2: Relational coordinates expose different organizations of the same analytical trajectory. **a**, Two analytically defined observables, $x_1(t)$ and $x_2(t)$, evolve over the same parameter t .¹ The target x_1 is colored by the sign of $\dot{x}_1$, with green indicating $\dot{x}_1 > 0$ and blue $\dot{x}_1 < 0$; open circles mark exact turning points $\dot{x}_1 = 0$. The reference x_2 is shown in purple, with dashed guides indicating its simultaneous value. **b**, The level-rate coordinates $(x_2, \dot{x}_2)$ leave the target branches strongly folded. **c**, The sensitivity-centered reference-shifted phase coordinates $(\dot{x}_2, D_{x_2}^s x_1)$ reorganize the same samples around the exact turning manifold $\dot{x}_1 = 0$ (black); the dashed green line marks the conditioning locus $g = \bar{g}$, where direct numerator sensitivity vanishes. **d**, The rate-rate coordinates $(\dot{x}_2, \dot{x}_1)$ place the same turning events on $\dot{x}_1 = 0$ while retaining reference-rate context. **e**, The rate-acceleration coordinates $(\ddot{x}_2, \dot{x}_1)$ place those events on the same manifold, indexed instead by local curvature. Across panels **b–e**, color denotes the same target branch, illustrating how alternative admissible coordinates in $\mathfrak{C}_q$ can expose different aspects of relational organization.

This example also illustrates why SIR does not posit a universally privileged coordinate family. The rate-rate representation makes the turning manifold particularly explicit; the rate-acceleration representation indexes the same events by local curvature; the trajectory-relational construction can organize the target relative to the evolution of another observable; and the original level-rate coordinates retain information that the alternatives transform or suppress. Which of these differences matters depends on q . Under one contract, branch separation or compact organization may justify promoting a construction within the search priority π_q ; under another, derivative uncertainty, conditioning, interpretability, computational cost, or deployment availability may make the same construction unattractive or inadmissible. Scientific precedent or prior qualified results may guide which coordinate families are examined first, but such proposal priority is not evidence that a representation is correct or well-suited for the task contract.

More generally, object-adapted coordinates enlarge the scientifically reachable relational universe by allowing the representation itself to participate in discovery. Their value lies not in mathematical novelty for its own sake, but in whether they preserve, expose, or simplify distinctions material to the declared question. The same freedom creates corresponding evidential obligations: constructed coordinates must retain their

domains, uncertainty, numerical realizations, information dependencies and provenance, and target-containing coordinates must be distinguished from independent predictive information. Accordingly, a coordinate system enters $\mathfrak{C}_q$ because it is admissible, enters the explored frontier because it is reached by the search policy, and survives as part of a qualified representation only when its utility is supported under $\mathcal{V}_q$. Further object classes, recursive coordinate construction, and the mathematical and numerical properties of trajectory-relational coordinates are developed in Supplementary Notes S2–S4.

1.3 Controlled Studies Isolate Distinct Qualified Discovery Outcomes

The framework of Section 1.1 does not imply that every discovery should terminate in a single equation. The form of the qualified output depends on the mathematical object X_q , the discovery contract $\mathcal{K}_q$, and what the available observations can distinguish within the explored admissible frontier. Thus we use controlled examples to isolate three consequences of this architecture. First, an ensemble object can support shared relational structure while retaining genuine realization-level variability. Second, changing the scientific task while holding the stored observations fixed can change which representation is preferred. Third, when several admissible relations remain observationally indistinguishable, the scientifically justified output is an ambiguity class rather than an arbitrarily selected generator. These studies are consequently probes of different stages of the SIR discovery hierarchy rather than benchmarks of unrelated fitting problems.

These controlled studies do not attempt open-ended ontology induction. Instead, they instantiate deliberately restricted and auditable versions of the general construction in Section 1.1. For each scientific object, Γ supplies a small object-appropriate ontology $\mathcal{G}_q$ containing only the coordinate constructors and relational templates needed to isolate the question under study; the corresponding search policy Σ_q is likewise kept simple, using finite enumeration or sparse structural selection rather than a learned or corpus-informed proposal mechanism. The ontology and search specification are frozen before protected evaluation. This restriction is intentional: it allows differences in the resulting $\mathcal{Q}_q^*$ to be attributed to the mathematical object, discovery contract and observational information rather than to an opaque or adaptive proposal process. More general ontology construction and search policies remain instances of the same architecture.

The first case changes the mathematical object supplied to the relational search. Consider the family $h_\alpha(x) = \sin(\alpha\pi x)$, with α sampled uniformly from $[1, 2]$. Treating the realizations jointly as an ensemble rather than as unrelated trajectories makes shared structure itself part of the discovery question. Each realization satisfies

$$a_i h_i'' + b_i h_i = 0, \quad \hat{\alpha}_i = \frac{\sqrt{b_i/a_i}}{\pi}.$$

Under a contract that admits common relational support with realization-dependent coefficients, SIR recovers the common two-term structure together with a distribution of inferred $\hat{\alpha}_i$ closely matching the sampled population (RMSE $\approx 1.6 \times 10^{-5}$, dominated by finite-difference boundary error). The qualified result is therefore not one coefficient vector averaged across the ensemble, but a shared relation together with a realization-level parameter population. This distinction becomes important whenever structural regularity persists across heterogeneous realizations even though their fitted parameters do not.

The pendulum study instead holds the stored observational object fixed and changes the discovery contract (Fig. 3a). The same 32-realization ensemble is interrogated first under a descriptive/compression contract q_{comp} . In this case, the useful representation is a shared invariant,

$$\omega^2 = -3.645(1 - \cos \theta) + 2E_i,$$

where E_i is the realization-specific conserved energy. The representation compresses each trajectory into a common geometric relation plus one scalar per realization. Under an evolution/prediction contract q_{pred} , however, that invariant is not the required scientific object. The admissible information, utility and validation requirements instead favor the evolution law

$$\dot{\theta} = \omega, \quad \dot{\omega} = -1.82245149 \sin \theta,$$

whose true second coefficient is -1.8225 . After the relation and its numerical realization are frozen, autonomous rollout from only the initial state of eight protected trajectories gives 30-s pooled RMSEs of 3.28×10^{-4} rad in θ and 3.38×10^{-4} rad s $^{-1}$ in ω .

The contrast is not that one pendulum representation is more correct than the other. The stored ensemble and underlying dynamics are unchanged, but $\mathcal{K}_{q_{\text{comp}}}$ and $\mathcal{K}_{q_{\text{pred}}}$ ask different scientific questions and impose different information and validation requirements. They therefore induce different useful regions of relational space and different qualified outputs. The compression contract favors a representation that exposes conserved geometric organization; the prediction contract favors a representation that can be propagated autonomously beyond the available information boundary. This is the operational meaning of task-conditioned representation in SIR: there need not be a task-independent C^* or R^* for the same observational record.

The final controlled study changes neither the optimizer nor the scientific task so much as the amount of discrimination supplied by the observational object. Consider the single field realization

$$u(x, t) = e^{-\alpha t} \sin(m\pi x), \quad \alpha = (m\pi)^2 \kappa.$$

It simultaneously satisfies

$$u_t + \alpha u = 0, \quad u_{xx} + (m\pi)^2 u = 0, \quad u_t - \kappa u_{xx} = 0.$$

The temporal eigen-relation, spatial eigen-relation and coupled diffusion relation are mathematically different statements, yet all are exact on the analytical realization and evaluate to $\|R\|_\infty < 10^{-4}$ under the finite-difference construction used here (Fig. 3b). The supplied field therefore does not contain enough variation to identify the heat equation uniquely from residual fidelity alone. Selecting the coupled relation merely because it is known to be the generator would import information that the discovery contract has not obtained from the observational object.

In the language of Section 1.1, several candidates in the explored admissible frontier consequently remain equivalent at the resolution required by the task. SIR therefore retains the supported relations as a qualified ambiguity class rather than promoting one member to a unique law. This is a positive scientific result: it identifies what the existing observations fail to determine and thereby constrains what additional evidence is needed. Supplementary Note S6 shows, for example, that observing multiple spatial modes under a shared-coefficient contract introduces variation that distinguishes the coupled diffusion relation from the mode-specific temporal and spatial eigen-relations. Formal distinctions among relational, observational and task equivalence are developed in Supplementary Note S5.

Together, these controlled studies demonstrate three different ways in which the qualified result $\mathcal{Q}_q^*$ can depart from the conventional notion of a single fitted equation. The mathematical interpretation of the record can make the relevant discovery a shared structure with a parameter population; the discovery contract can select different representations of the same stored observations; and limited observational discrimination can require that several surviving representations remain unresolved. Conventional single-generator recovery is recovered when the representation and relation family are sufficiently identifying, and is considered directly in the restricted-model setting of the following section. The broader principle is that SIR returns the strongest relational statement supported by the object, contract and validation evidence, rather than forcing every discovery into the same output type.

1.4 Conventional Model Identification Is Nested Within Relational Discovery

PySINDy solves a model-identification problem after much of the scientific problem formulation has been supplied. SIR makes that problem formulation—the scientific object, information boundary, admissible coordinates, relational families, utility and validation requirements—an explicit computational object that participates in discovery. This distinction does not require a competing numerical solver: PySINDy can itself be used as a relation estimator within a SIR contract.

In the language of Section 1.1, conventional system identification corresponds to a deliberately restricted region of the SIR architecture. The task-admissible record $\mathcal{O}_q$ and its mathematical interpretation X_q are

Same Trajectories, Different Task Contracts

Same 32-Realization Observational Ensemble

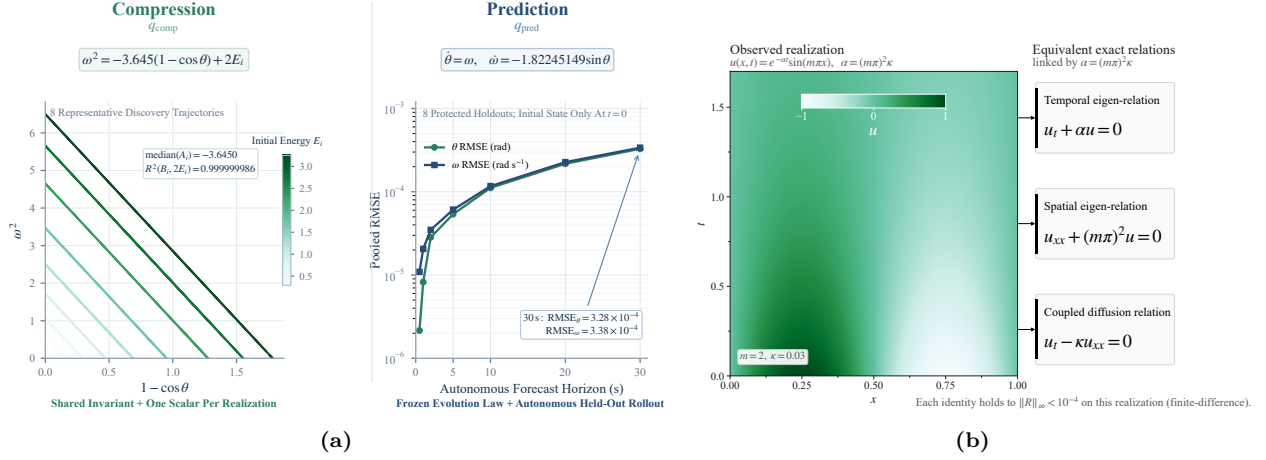


Figure 3: The discovery contract and observational support determine the qualified representation. a, The same 32-realization nonlinear-pendulum ensemble supports different representations under different scientific contracts. A compression contract recovers the shared invariant $\omega^2 = -3.645(1 - \cos \theta) + 2E_i$, reducing each realization to a common relation and its conserved-energy parameter. A prediction contract instead recovers the evolution law $\dot{\theta} = \omega, \dot{\omega} = -1.82245149 \sin \theta$. With the predictive representation frozen, autonomous rollouts on eight protected trajectories using only their initial states achieve 30-s pooled RMSEs of 3.28×10^{-4} rad in θ and 3.38×10^{-4} rad s^{-1} in ω . **b,** One separated heat-equation realization simultaneously supports temporal, spatial and coupled differential relations ($\|R\|_{\infty} < 10^{-4}$ under the displayed finite-difference evaluation). The observations therefore support an ambiguity class rather than identifying the generating PDE uniquely. Together, the panels illustrate complementary consequences of the SIR framework: changing $\mathcal{K}_q$ can change the preferred representation even when the stored observations are unchanged, whereas insufficient discrimination within a fixed observational object limits the strength of the qualified claim.

supplied in advance; the ontology $\mathcal{G}_q$ is restricted to a prescribed coordinate system and model grammar; and the explored frontier $\hat{\mathfrak{A}}_q$ consequently contains only relations expressible within that representation. A sparse, symbolic, neural or other estimator can then instantiate the relation-search and fitting component of Σ_q . SIR differs not by requiring a different estimator, but by permitting these upstream representational choices to become explicit, searchable parts of the discovery problem when the contract allows them.

The conventional Lorenz problem provides the limiting case. For the generator recovery contract q_{gen} , the information operator exposes the full state (x, y, z) , $X_{q_{\text{gen}}}$ is a three-dimensional trajectory, and the ontology is intentionally restricted to the observed state, its temporal rates and a cubic polynomial relation family. Using $\sigma = 10$, $\rho = 28$ and $\beta = 8/3$, finite-difference derivatives and sparse structural identification recover

$$\dot{x} = -10.000x + 10.000y, \quad \dot{y} = 28.000x - 0.999y - 0.999xz, \quad \dot{z} = -2.667z + 1.000xy.$$

The canonical support is recovered with coefficient deviations of approximately 2×10^{-3} . Supplying manufactured exact derivatives while leaving the relation search unchanged preserves the support and reduces the remaining coefficient error (Supplementary Note S6). This study is therefore a containment test rather than evidence of superiority over conventional sparse identification: once C , $\mathfrak{A}_q(C)$ and the relevant numerical realizations are fixed, SIR reduces to the familiar model-identification problem.

We next enlarged the problem at the representational rather than numerical level. For the reconstruction contract q_{rec} , the underlying Lorenz system is unchanged, but the information boundary admits x and y as explanatory observations while z is available only as the supervised quantity to be reconstructed. The mathematical object is therefore a partially observed trajectory, and Γ instantiates a finite, auditable ontology of coordinate constructions obtainable from the permitted observations. Rather than fixing one coordinate system in advance, this ontology induces a family $\mathfrak{C}_{q_{\text{rec}}}$ of admissible representations, whose complete expanded member C_{all} contains 38 coordinates. PySINDy is retained as the common sparse relation

estimator throughout, so changes in performance can be attributed to the representation and discovery contract rather than to a change of regression algorithm.

Representation search and selection were performed using 48 development trajectories and frozen before evaluation on 24 independently generated trajectories. Within the same clean observational object and admissible relational universe, an accuracy-focused contract selected $C_{\text{acc}}^* = C_{\text{all}}$, giving a held-out RMSE of 1.35×10^{-5} . A compactness-focused contract instead selected a 12-coordinate representation C_{comp}^* , with held-out RMSE 3.26×10^{-5} . Both are effectively exact on the scale of the Lorenz state, despite differing by more than a factor of three in coordinate count. A conventional quadratic polynomial representation constructed from the same raw observational information gave RMSE 2.27.

This comparison separates two roles that are often conflated. PySINDy determines which sparse relation is useful *within* a supplied coordinate representation; SIR additionally asks which admissible coordinate representation should be supplied. Indeed, the full expanded representation C_{all} was itself passed directly to the same PySINDy estimator, demonstrating that no special regression mechanism is required to exploit the additional coordinates once they have been constructed. The substantive difference lies upstream: the coordinate grammar, admissibility conditions, representation search and task-conditioned comparison are made explicit parts of the scientific problem rather than fixed preprocessing assumptions.

We then changed the observational conditions while retaining this finite representation experiment. Under 1% observational noise, the compact 12-coordinate representation was substantially more fragile, with RMSE 5.12, whereas a robustness-focused contract retained the complete representation, which gave RMSE 2.25; the conventional quadratic representation gave RMSE 3.00. An unchanged multilayer perceptron showed the same qualitative distinction: the compact relational representation provided a strong advantage for the clean observations but not once noise was introduced. Thus even within one fixed admissible coordinate grammar, observational uncertainty can change which region of $\mathfrak{C}_q$ is useful under the declared contract.

This noise experiment is deliberately narrower than the full ontology-refinement mechanism of Section 1.1. To isolate robustness of the candidate representations already under consideration, the coordinate-construction grammar and numerical realization family were held fixed rather than allowing Γ to rebuild the ontology in response to the new error model. A fully adaptive SIR analysis could additionally change derivative estimators, smoothing or regularization procedures, admissibility thresholds or other numerical realizations when the uncertainty structure changes. The present comparison therefore tests task-conditioned selection within a controlled ontology, not the stronger claim that the displayed noisy representation is the best one obtainable after ontology refinement.

Taken together, the two Lorenz studies expose a nested hierarchy. Fixing the scientific object, coordinate system and relation family reduces SIR to conventional model identification. Holding the estimator fixed while enlarging $\mathcal{G}_q$ and $\mathfrak{C}_q$ converts the problem into representation discovery. Changing U_q , the information boundary or the observational error model can then change which member of the same admissible universe is retained in $\mathcal{Q}_q^*$. These results are therefore not evidence that PySINDy, neural networks or other conventional estimators are intrinsically incapable of using relational representations. Rather, they demonstrate the distinction between solving for a model after a representation has been supplied and treating the construction, exploration and qualification of that representation as part of scientific discovery itself.

1.5 Multimodal Plasma Observations Support Distinct Task-Conditioned Discoveries

We next asked whether the architecture of Section 1.1 remains useful when the scientific object is neither synthetic nor naturally supplied in a privileged mathematical representation. We treated 62 DIII-D tokamak discharges as a common provenance-bearing observational system containing eight plasma quantities,

$$\{W_{\text{dia}}, P_{\text{NBI}}, n_e, I_p, \beta_N, q_{95}, \ell_i, \kappa\},$$

where W_{dia} is the diamagnetic stored-energy estimate, P_{NBI} the injected neutral-beam power, n_e the electron density, I_p the plasma current, β_N the normalized plasma beta, q_{95} the safety factor at the 95% poloidal-flux

surface, ℓ_i the internal inductance, and κ the plasma elongation. These quantities span energetic state, external actuation, density, current, stability and reconstructed equilibrium geometry, but they are not homogeneous observations: they differ in physical meaning, provenance, sampling and reconstruction history. The 62 discharges are therefore treated as an ensemble of multimodal trajectories rather than as one undifferentiated numerical array. Accordingly, the task-conditioned mathematical object X_q retains both the within-discharge temporal structure and the realization index across the discharge ensemble.

For this experimental study we did not attempt open-ended ontology induction. Instead, Γ instantiated a finite, auditable relational ontology appropriate to the trajectory-ensemble object. Its vocabulary included the admissible primitive observables together with temporal-rate coordinates, algebraic ratios and first-order trajectory-relational coordinates, with availability, denominator and provenance requirements inherited from the discovery contract. Sparse structural selection then provided the search policy Σ_q over the resulting finite explored frontier. Thus, as in the controlled studies, the ontology and search policy are restricted instantiations of the general architecture rather than assumptions that these coordinate families exhaust the scientifically meaningful relational universe.

The same stored observational system was interrogated under two different discovery contracts (Fig. 4). Under the descriptive contract q_{desc} , the objective was to identify compact relational organization within the realized plasma trajectories. Contemporaneous quantities and target-containing coordinates were therefore admissible because the task did not require independent prediction of an unavailable response. Under the reconstruction contract q_{rec} , by contrast, the designated target was I_p , and I_p together with all target-dependent descendants was excluded from the explanatory coordinate grammar. The two contracts therefore act on the same underlying scientific object but induce different admissibility conditions, explored relational frontiers and standards of qualification.

Within the descriptive contract, sparse search over the admissible frontier identified the common seven-coordinate support

$$C_{\text{desc}}^* = \left\{ D_{\kappa} W_{\text{dia}}, D_{\beta_N} W_{\text{dia}}, D_{\beta_N} \kappa, D_{\beta_N} \ell_i, \frac{q_{95}}{\kappa}, \dot{\beta}_N, \dot{\kappa} \right\}.$$

The selected representation mixes three forms of organization that would not appear together in a raw-variable model: relative trajectory evolution through phase-derivative coordinates, explicit temporal evolution through $\dot{\beta}_N$ and $\dot{\kappa}$, and algebraic structure through q_{95}/κ . The representation therefore illustrates directly the role of $\mathcal{G}_q$: the selected relation is expressible only because the ontology makes these heterogeneous coordinate constructions jointly reachable.

The seven-coordinate support is shared across the 62-discharge ensemble, while its coefficients are estimated separately within each discharge. With those realization-specific coefficients, the pooled reconstruction RMSE is 5.76×10^{-2} in standardized target coordinates. As in the controlled ensemble example of the preceding section, the qualified object is consequently not one universal coefficient vector. It is a common relational architecture together with a population of realization-specific coefficient vectors. The cohort-mean coefficients shown in Fig. 4a summarize that population rather than defining a single equation applied uniformly to every discharge.

The trajectory-relational coordinates also give the descriptive result a useful geometric interpretation. Because $D_g f = \dot{f}/\dot{g}$ measures the local evolution of one observable relative to another along the realized trajectory, the selected support can be interrogated as a map of *observational relational sensitivity*: it identifies combinations of plasma quantities whose local evolution is strongly organized with the realized response. Such structure can prioritize variables, regimes or candidate relations for subsequent physical investigation. It does not, however, assign causal direction. The phase derivatives are trajectory-conditioned observational relations, not causal response coefficients.

This distinction is especially important because $D_{\kappa} W_{\text{dia}}$ and $D_{\beta_N} W_{\text{dia}}$ contain the contemporaneous response derivative. Their admissibility follows from $\mathcal{K}_{q_{\text{desc}}}$: retrospective description permits information that an independent predictor could not use. The resulting coordinate–relation pair is therefore a target-containing implicit closure, not an independent predictor of $\dot{W}_{\text{dia}}$. Its scientific content is the compact relational organization supported by the realized trajectories. Algebraic elimination, coefficient identifiability, denominator conditioning and target-map conditioning further qualify this interpretation in Supplementary

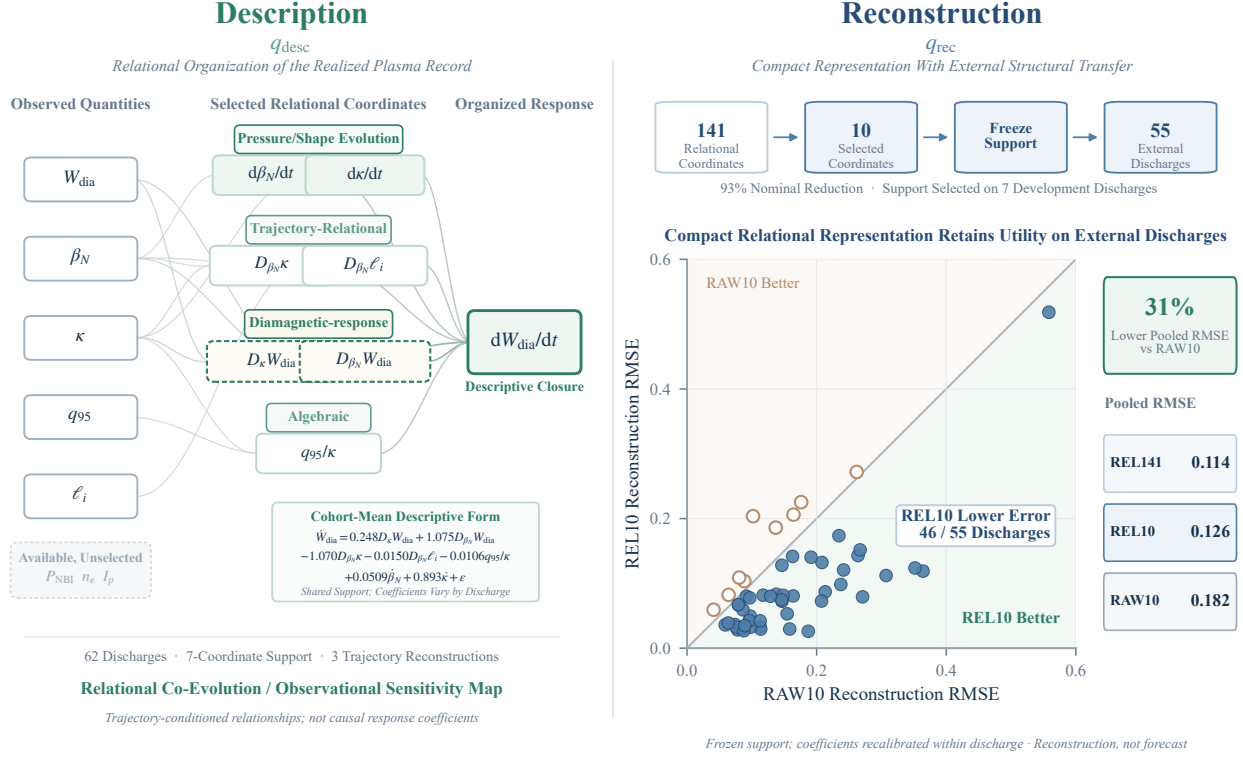


Figure 4: Distinct discovery contracts induce different qualified representations of the same multimodal DIII-D observational system. a, Under the descriptive contract q_{desc} , the task-admissible object is treated as an ensemble of multimodal plasma trajectories and the restricted relational ontology admits temporal, algebraic and trajectory-relational coordinate constructions. SIR selects a seven-coordinate organization of standardized dW_{dia}/dt across 62 discharges. Target-containing coordinates are explicitly marked because they are admissible for retrospective description but not for independent prediction. The same selected organization is examined under aligned, spline and Rauch–Tung–Striebel trajectory realizations. **b,** Under the reconstruction contract q_{rec} , I_p and target-dependent descendants are excluded from the explanatory coordinate grammar. A ten-coordinate support is selected from an explored frontier of 141 relational coordinates using seven development discharges and then frozen before evaluation on 55 external discharges. The compact relational representation (REL10) achieves pooled reconstruction RMSE 0.126, compared with 0.182 for a size-matched raw-coordinate representation (RAW10), and has lower error on 46 of 55 external discharges. The dense 141-coordinate relational reference achieves RMSE 0.114. Only the coordinate support is transferred; coefficients are calibrated within each external discharge.

Note S7.

The same scientific coordinate also admits more than one numerical realization. We therefore evaluated the dominant descriptive organization using aligned signals, spline reconstructions and Rauch–Tung–Striebel trajectory reconstructions. These variants do not change the semantic meaning of a rate or phase-derivative coordinate; they alter the numerical procedure by which that coordinate is realized from a finite experimental record. Persistence of the selected organization across these admissible realizations consequently tests the distinction made in Section 1.1 between a scientific coordinate and its numerical construction. Complete reconstruction and derivative provenance is given in Supplementary Note S7.

We then changed the discovery contract rather than attempting to promote the descriptive closure to a stronger claim. Under q_{rec} , the scientific question was whether a compact representation of the admissible plasma observations could reconstruct I_p . The target and all target-dependent descendants were excluded before coordinate construction, so Γ induced a different task-admissible coordinate space. The resulting explored frontier contained 141 relational coordinates. Seven development discharges were used by $\Sigma_{q_{rec}}$ to select a ten-coordinate support, after which the representation was frozen before the remaining 55 discharges were examined.

The external discharges test transfer of the discovered representation rather than transfer of one universal coefficient vector. Within each external discharge, coefficients were calibrated on the first 80% of the trajectory and reconstruction was evaluated on the protected final 20%. The resulting REL10 representation reduced the nominal relational basis from 141 to 10 coordinates while retaining most of the performance of the dense relational reference: pooled reconstruction RMSE was 0.126 for REL10 and 0.114 for REL141. At matched dimensionality, the raw-coordinate representation RAW10 gave RMSE 0.182, with REL10 obtaining lower error on 46 of 55 external discharges (Fig. 4b). The effective degrees of freedom similarly decreased from approximately 99 to 10.6.

The result is therefore not primarily that I_p can be reconstructed, which is deliberately an ordinary task. Rather, the ontology makes a larger relational universe reachable, the task-conditioned search identifies a much smaller coordinate system within that universe, and the utility of that coordinate system survives structural transfer to discharges excluded from support selection. In the language of Section 1.1, the transferred scientific object is principally the selected representation C_{rec}^* ; its numerical coefficients remain realization-specific.

The two plasma contracts consequently instantiate different paths through the same SIR architecture. For q_{desc} , the information boundary admits contemporaneous relational coordinates and the qualified output is a shared descriptive organization with realization-dependent coefficients. For q_{rec} , target containment is prohibited, the ontology and explored frontier change accordingly, and the qualified output is a compact coordinate support whose structural utility transfers to external discharges. Neither result is presented as a universal causal plasma law or a prospective forecast. They are different scientific answers because the questions, information boundaries and validation requirements are different.

This experimental study also illustrates why the distinctions introduced in Section 1.1 matter in practice. The primitive record does not uniquely determine the mathematical representation; the ontology determines which relational hypotheses can become expressible; the contract determines which of those are scientifically admissible; the search policy determines which portion of that space is actually explored; and validation determines the strength of the claim that can travel with the selected representation. The output of the analysis is therefore not simply a low-residual plasma equation, but a task-conditioned Q_q^* whose coordinates, domain, provenance, information availability, validation evidence and limitations remain attached to the discovery.

2 Discussion

SIR changes the computational unit of scientific discovery from an equation written in predetermined variables to a *qualified representation*. In the framework developed here, a scientific question acts on a provenance-bearing observational object through an explicit information boundary; the resulting record is assigned a task-conditioned mathematical interpretation, from which a relational ontology determines what coordinate and relational constructions are scientifically meaningful. A search policy explores a finite portion of that admissible universe, and validation determines which coordinate–relation pairs can survive as qualified scientific outputs. The result is therefore not only R^* , but a task-conditioned object Q_q^* whose coordinates, domain, provenance, information availability, evidential support and unresolved limitations remain attached to the claim.

This separation matters because representation is itself part of scientific reasoning. Many successful mathematical developments amount to finding coordinates in which a difficult organization becomes simpler: nonlinear structure may become approximately linear, folded branches may become separable, a field may become sparse in a modal basis, or a complicated trajectory may become organized around an invariant or low-complexity manifold. The trajectory-relational examples in this work illustrate this principle rather than defining its scope. Phase derivatives compare the local evolution of observables along a common trajectory, while level–rate, rate–rate and rate–acceleration coordinates interrogate complementary aspects of the same record. In the controlled example, a trajectory-relational construction reorganizes a strongly folded target around an explicitly characterized turning manifold. More generally, a useful coordinate can trade representational complexity for relational simplicity. Whether that trade is scientifically useful depends

on the task, numerical realization, uncertainty and information boundary rather than on the mathematical novelty of the coordinate itself.

The distinction between a *relational ontology* and the search performed within it is equally important. SIR does not require an exhaustive enumeration of all mathematically expressible hypotheses. The ontology $\mathcal{G}_q$ specifies the task-conditioned grammar of reachable scientific constructions, whereas the search policy Σ_q determines how finite computational effort is allocated within the resulting admissible universe. Consequently, the explored frontier $\hat{\mathfrak{A}}_q$ is generally only a subset of the admissible space $\mathfrak{A}_q$. A candidate absent from a discovery result may therefore be inadmissible, may have been evaluated and rejected, or may simply not have been reached. This distinction places an important boundary on negative scientific claims: failure to discover a relation is conditional on the ontology, numerical realizations, search policy and computational bounds under which the search was conducted.

The implementations in this Article deliberately exercise restricted versions of this general architecture. Their ontologies are finite and transparent, and their search policies use constrained enumeration, sparse regression or greedy structural selection rather than attempting open-ended automated ontology construction. This choice makes the controlled experiments interpretable. The ensemble example isolates the consequence of treating repeated observations as one mathematical object and returns common relational support together with a population of realization-specific parameters. The pendulum example holds the stored observations fixed while changing the discovery contract and therefore returns an invariant under a compression task and an autonomous evolution law under a prediction task. The heat-field example demonstrates the complementary case in which the observational object cannot distinguish several exact relations, so the justified result is an ambiguity class rather than an arbitrarily selected generator. These are not progressively better or worse versions of equation recovery; they are different forms of $\mathcal{Q}_q^*$ justified by different combinations of object, question and evidence.

Conventional model identification occupies a limiting case of the same construction. When the information boundary, coordinate system, model grammar and numerical realizations are fixed in advance, SIR reduces to search over the remaining relation parameters and structures. The full-state Lorenz study therefore recovers conventional sparse generator identification rather than a new competing model class. Expanding the admissible coordinate ontology while retaining PySINDy as the relation estimator changes the scientific problem: the algorithm is no longer asked only which sparse relation fits a supplied representation, but also which admissible representation should be supplied to that estimator. Accuracy- and compactness-focused contracts then select different Lorenz coordinate systems from the same clean observations. The noise study further shows that observational conditions can alter the preferred representation, while also exposing an important limitation of the present benchmark: its coordinate grammar and numerical realization family were held fixed. A fully adaptive SIR analysis could revise derivative estimators, smoothing, regularization or other coordinate realizations when the uncertainty model changes. Thus, the general framework is broader than the particular controlled search exercised here.

The DIII-D application shows why these distinctions become consequential for a heterogeneous experimental object. The 62-discharge record combines quantities with different physical roles, sampling histories and reconstruction dependencies, and does not supply a uniquely privileged representation. Under the retrospective descriptive contract, the admissible ontology includes target-containing trajectory-relational coordinates and yields a common seven-coordinate organization with realization-dependent coefficients. That representation combines relative trajectory evolution, explicit temporal evolution and algebraic structure, and remains qualitatively organized across distinct trajectory reconstructions. Because two selected coordinates contain the contemporaneous response derivative, however, the result is properly qualified as a target-containing implicit closure rather than an independent predictor. Coefficient identifiability, algebraic constraint content and denominator and target-map conditioning further limit the strength of any parameter-level interpretation (Supplementary Note S7). These qualifications are not ancillary caveats; under SIR they are part of the scientific object returned by the discovery.

Changing the DIII-D contract changes the admissible ontology before model selection. For reconstruction of I_p , the target and its descendants are excluded from the explanatory coordinate grammar. From an explored frontier of 141 relational coordinates, support selection on seven development discharges produces

a ten-coordinate representation that is frozen before structural evaluation on 55 external discharges. The compact representation retains most of the dense relational performance and outperforms a size-matched raw representation on 46 of the 55 external discharges. What transfers here is the coordinate support, not a universal coefficient vector: coefficients are recalibrated within each discharge, and the protected segment is a reconstruction holdout rather than a prospective forecast. The justified claim is therefore narrower, but also more informative, than simply stating that a plasma variable can be fitted. A relational coordinate system selected on one subset of experimental realizations retains useful structure on discharges excluded from its selection.

The broader implication is that identifiability is conditional not only on a model class, but on the observational object and discovery contract. A finite record may identify a generator, a shared structural family, a task-specific representation, an equivalence class, or no useful representation at the required evidential level. No transformation can recover distinctions absent from every admissible observation, and more powerful optimization cannot resolve observational equivalence. When several qualified representations remain indistinguishable, preserving that ambiguity and asking which new measurement, perturbation, parameter variation or intervention would separate the alternatives is a stronger scientific response than selecting one expression because an optimizer assigns it a marginally better score.

The framework also suggests a route toward more capable automated discovery without making automated proposal synonymous with scientific evidence. External scientific knowledge $\mathcal{H}_q$, including established theory, peer-reviewed literature, curated databases, laboratory records or previous qualified discoveries, can inform a proposal priority π_q that directs search toward promising regions of relational space. Learned models, scientific agents, simulators and existing numerical codes can similarly act as typed constructors or search mechanisms within the ontology. Such mechanisms can make discovery dramatically less combinatorial, but their role is to determine *where to look*, not *what to believe*. Search precedent is not validation evidence, and low-priority admissible regions must remain accessible when the purpose of the search includes genuinely novel representations. Provenance of the knowledge source, search policy and explored frontier is therefore necessary if such systems are to remain auditable.

Several limitations follow directly from this architecture. No finite ontology is complete, and the particular ontologies instantiated here cannot establish that better representations do not exist outside their explored frontiers. Coordinate construction can amplify noise, introduce singularities or create algebraic dependence; learned or model-derived coordinates can inherit biases and uncertainty from upstream procedures. Greedy and sparse searches need not find globally preferred structures. Task-conditioned utility can itself be misspecified, and an information boundary that is appropriate for retrospective description may be invalid for prediction or control. Prospective and causal claims therefore require stronger contracts than those exercised by most of the present studies: every coordinate must be available at the relevant decision time, structural choices must be frozen before protected evaluation, and generalization, intervention or closed-loop behavior must be tested at the level actually being claimed.

These limitations motivate the central discipline of SIR rather than weakening it. Relational discovery is not intended to make every mathematical construction scientifically acceptable or to replace theory with a larger feature search. Its purpose is to make explicit the sequence of assumptions that normally precedes model fitting: what the observational object is, what information may be used, which scientific constructions are meaningful/admissible, which portion of that space was explored, what utility defined success, and what evidence supports the final interpretation. In this view, the strongest discovery is not necessarily the simplest equation, the most accurate predictor, or the most theoretically rigorous construction. It is the strongest qualified relational statement that the available observational system and declared scientific contract can support.

Methods

Task-conditioned study design and contract instantiation

Each analysis was organized as an instantiation of the task-conditioned discovery architecture defined in Section 1.1. For a provenance-bearing scientific object $\mathcal{O}$, a scientific question q specified a discovery contract

$$\mathcal{K}_q = (q, \mathcal{I}_q, \mathcal{P}_q, \mathcal{B}_q, \mathcal{H}_q, U_q, \mathcal{V}_q, \Omega_q),$$

from which the information-admissible record $\mathcal{O}_q = \mathcal{I}_q(\mathcal{O})$ and its mathematical interpretation $X_q = \text{Obj}_q(\mathcal{O}_q)$ were defined. The contract specified the information boundary, semantic and numerical admissibility requirements, complexity and search bounds, task utility, validation protocol and intended domain of the resulting claim.

For the studies reported here, the general ontology-construction map

$$\mathcal{G}_q = \Gamma(X_q, \mathcal{O}_q, \mathcal{K}_q; \Lambda, \mathfrak{T})$$

was instantiated using finite and deliberately auditable coordinate and relation grammars. The controlled studies were designed to isolate particular properties of the framework and therefore used small object-appropriate ontologies. The DIII-D studies used a larger but still finite grammar comprising admitted primitive quantities and declared algebraic, differential and trajectory-relational constructions. No reported result depended on an automated literature-mining or learned proposal-priority system. Such procedures are admissible under the general framework but are prospective extensions rather than components of the present experiments.

The ontology induced an admissible coordinate–relation universe

$$\mathfrak{A}_q = \{(C, R) : C \in \mathfrak{C}_q, R \in \mathfrak{R}_q(C)\}.$$

The computational search did not assume exhaustive exploration of this universe. Instead, the instantiated search policy Σ_q , together with the complexity bounds in $\mathcal{B}_q$, generated an evaluated frontier

$$\widehat{\mathfrak{A}}_q = \text{Explore}(\mathfrak{A}_q; \Sigma_q, \mathcal{B}_q) \subseteq \mathfrak{A}_q.$$

Consequently, negative results are conditional on the declared ontology, numerical realizations, search policy and explored frontier.

Where protected evaluation was used, development and evaluation evidence were kept distinct. Ontology construction, coordinate selection, complexity choices and other adaptive decisions were performed using $\mathcal{O}_q^{\text{dev}}$; the resulting representation and search choices were frozen before examination of $\mathcal{O}_q^{\text{eval}}$. A protected observation that influenced a subsequent representational decision would cease to constitute protected evaluation and would instead be reclassified as development evidence.

Observational-object audit, ontology instantiation and coordinate realization

Before coordinate construction, each admitted record was audited as a scientific object rather than treated as an untyped numerical matrix. The audit recorded the source type; natural temporal, spatial or ensemble domain; realization indices; sampling and ordering conventions; masks and boundary rules; uncertainty and known reconstruction dependencies; information availability under the scientific task; and the provenance of transformations required to place quantities on a common evaluable support. Experimental quantities additionally retained whether they were directly measured, calibrated, processed, model-inferred or reconstructed from other measurements.

The mathematical interpretation X_q determined which constructors were eligible to participate in the restricted ontology. Depending on the object, the admitted vocabulary included recorded values, algebraic functions,

temporal or spatial derivatives, integrals, delays, trigonometric or dimensionless constructions, realization-level summaries and trajectory-relational coordinates. Relation templates included explicit regression, implicit constraints, shared-support families with realization-specific coefficients and, where applicable, rational or target-containing forms. Formal typing, composition and admissibility rules are given in Supplementary Notes S1–S3.

A scientific coordinate was distinguished from its numerical realization. For example, $\text{Rate}(x)$ denotes the semantic rate coordinate, whereas a finite-difference stencil, analytical derivative, spline derivative or smoothed-state derivative is a particular realization of that coordinate. The numerical realization therefore carries its own provenance, including the sampling scale, reconstruction method, derivative rule, boundary convention, standardization and validity mask. Distinct realizations of the same semantic coordinate were compared only when permitted by the corresponding discovery contract.

For differentiable observables f and g along a trajectory γ , the classical trajectory-relational derivative, referred to in this work as a phase derivative, was evaluated as

$$D_g^\gamma f = \frac{\partial_\gamma f}{\partial_\gamma g}$$

on the domain for which the numerator and denominator derivatives exist and $\partial_\gamma g \neq 0$. Sampled realizations additionally required a declared denominator-conditioning rule. Shifted, bounded, reference-shifted and sensitivity-centered constructions were treated as distinct coordinates rather than as numerically interchangeable estimators of the classical quotient. Their definitions, regularity, transformation properties and conditioning are developed in Supplementary Note S4.

The analytical turning-manifold example in Fig. 2 used the prescribed functions $x_1(t)$ and $x_2(t)$ reported with that figure. Their derivatives were evaluated analytically so that the comparison isolated coordinate geometry rather than differentiation error. The sensitivity-centered trajectory-relational construction and its exact turning manifold are specified in Supplementary Note S4.

Relation estimation, search policy and qualification

Conditional on an evaluated coordinate system C and relation structure R , numerical parameters were fitted using the estimator appropriate to the restricted relation family. For an explicit library $\Psi(C)$ and response $\mathbf{Y}$, this included problems of the form

$$\hat{\boldsymbol{\theta}} \in \arg \min_{\boldsymbol{\theta}} \mathcal{L}_q^{\text{fit}}(\mathbf{Y}, \Psi(C)\boldsymbol{\theta}),$$

possibly with sparsity or structural constraints. Homogeneous implicit relations were represented as

$$\Psi(C)\mathbf{a} \approx \mathbf{0},$$

with a normalization imposed to exclude the trivial coefficient vector. Shared-support ensemble models retained realization-specific coefficient vectors rather than automatically replacing them by their mean.

Search and coefficient estimation were treated as distinct operations. Depending on the study, Σ_q used finite comparison of predeclared representations, sparse structural estimation or greedy support selection with coefficient refitting. PySINDy was used as the common sparse relation estimator in the Lorenz nesting and representation-expansion studies; its role was to fit relations after a candidate coordinate representation had been supplied. The expanded representations were therefore supplied directly to the same estimator rather than compared against a privileged SIR-specific regression algorithm.

Candidate selection was performed under the task utility U_q and validation protocol $\mathcal{V}_q$. The utility was not assumed to be a universal scalar model score. Depending on the contract, candidates were compared by residual or reconstruction fidelity, structural recovery, parameter recovery, compactness, stability, held-out transfer, autonomous rollout performance or combinations of these criteria. Candidates that remained indistinguishable within the declared task resolution could be retained as a qualified equivalence class rather than forced into a unique ranking.

Target containment was evaluated separately from fit quality. A coordinate containing the contemporaneous response was admissible only when permitted by the information boundary. Such a coordinate can participate in a descriptive implicit constraint but cannot provide independent predictive information about that same response. Algebraic dependence, local constraint content, denominator support and target-map conditioning were evaluated where relevant; the formal tests and their DIII-D application are given in Supplementary Notes S5 and S7.

Controlled-study protocols

Conventional Lorenz generator recovery. The Lorenz system was evaluated with $\sigma = 10$, $\rho = 28$ and $\beta = 8/3$. For the conventional generator contract, the full state (x, y, z) was admitted and the ontology was restricted to state coordinates, their temporal rates and monomials through cubic order. The canonical trajectory was initialized from $(1, 1, 1)$. Three-point central differences using the physical temporal spacing provided the primary numerical rate realization. Manufactured exact derivatives were supplied in a separate derivative-fidelity ablation while leaving the relation search unchanged. Support and coefficient recovery were compared with the known generator. Additional derivative-fidelity analysis is given in Supplementary Note S6.

Partially observed Lorenz representation discovery. For the representation-expansion study, the underlying Lorenz dynamics were unchanged but the scientific question and information boundary were altered. The explanatory observations were restricted to x and y , while z was retained only as the supervised reconstruction target. A finite ontology of object-admissible local, differential, algebraic and trajectory-relational constructions was instantiated from the permitted observations. Seventeen predeclared representation families were compared; their union formed the 38-coordinate representation denoted C_{all} . The complete coordinate inventory and representation families are given in Supplementary Note S6.

Representation selection used 48 development trajectories and was frozen before evaluation on 24 independently generated confirmation trajectories. PySINDy with the same sparse-estimation configuration was used for every candidate representation. The accuracy contract prioritized held-out reconstruction fidelity. The compactness contract imposed a practical fidelity-equivalence threshold before preferring a lower-dimensional representation. A conventional quadratic polynomial representation generated from the same admitted raw observations was retained as a baseline.

For the robustness ablation, the same finite coordinate grammar and estimator family were retained while the admitted observations were perturbed at the predeclared 1% noise level. This experiment therefore tests robustness within a fixed ontology rather than ontology reconstruction under a changed error model. The perturbation generator, random state and complete candidate definitions are part of the frozen benchmark provenance reported in Supplementary Note S6. A multilayer perceptron with unchanged architecture and training protocol across coordinate representations was used as an estimator ablation to separate representational effects from sparse-regression-specific effects.

Random-coefficient oscillator ensemble. For

$$h_\alpha(x) = \sin(\alpha\pi x), \quad \alpha \sim U[1, 2],$$

40 independent realizations of α were evaluated on a common 100-point spatial grid. The exact second derivative was supplied in the canonical analysis so that the experiment isolated shared-support and coefficient-population recovery rather than numerical differentiation error. The restricted ontology admitted h , its spatial derivatives and an implicit two-term relation with support shared across the ensemble and coefficients fitted separately by realization. The inferred scale-invariant coefficient ratio was transformed to $\hat{\alpha}_i$ and compared with the sampled population. The empirical parameter-distribution error reported in Results is therefore a realization-level recovery metric rather than a pooled single-equation fit.

Pendulum task-conditioning study. The pendulum experiment comprised 32 trajectory realizations of the same nonlinear dynamical system. The mathematical object was held fixed while two discovery contracts were applied. The restricted ontology admitted the state coordinates (θ, ω) , object-appropriate trigonometric, quadratic-energy and temporal-rate constructions.

Under q_{comp} , the utility favored a common low-dimensional description of the trajectory ensemble and permitted a realization-specific scalar parameter. The selected representation was the shared invariant

$$\omega^2 = -3.645(1 - \cos \theta) + 2E_i,$$

where E_i is the realization-specific conserved energy. Under q_{pred} , the information and validation contract instead required an autonomously propagatable evolution law and selected

$$\dot{\theta} = \omega, \quad \dot{\omega} = -1.82245149 \sin \theta.$$

Twenty-four trajectories were used for development and structural selection. The representation was then frozen and the remaining eight trajectories were used as protected rollout cases. Only the initial state at $t = 0$ was supplied for the protected autonomous rollout; pooled errors were evaluated over the 30-s forecast horizon. Further contract and validation details are given in Supplementary Note S6.

Heat-field identifiability study. The field

$$u(x, t) = e^{-\alpha t} \sin(m\pi x), \quad \alpha = (m\pi)^2 \kappa,$$

was treated as a single spatiotemporal object. Analytical u_t and u_{xx} were supplied so that the test isolated representational identifiability. The restricted field ontology admitted temporal and spatial differentiation and contained the temporal eigen-relation, spatial eigen-relation and coupled diffusion relation

$$u_t + \alpha u = 0, \quad u_{xx} + (m\pi)^2 u = 0, \quad u_t - \kappa u_{xx} = 0.$$

Residual and zero-set equivalence were evaluated on their common supported domain. Because the single separated realization supports all three relations, the corresponding qualified result retains the ambiguity rather than selecting the known generating PDE from external knowledge. Supplementary Note S6 examines how additional mode variation can reduce this ambiguity under a shared-coefficient contract.

DIII-D observational object and preprocessing

The plasma study used 62 DIII-D discharge realizations containing eight admitted quantities,

$$\{W_{\text{dia}}, P_{\text{NBI}}, n_e, I_p, \beta_N, q_{95}, \ell_i, \kappa\}.$$

Here W_{dia} is the diamagnetic stored-energy estimate, P_{NBI} the injected neutral-beam power, n_e the electron density, I_p the plasma current, β_N the normalized plasma beta, q_{95} the safety factor at the 95% poloidal-flux surface, ℓ_i the internal inductance and κ the plasma elongation. The corresponding source identifiers were `pcdiamag3`, `pinj`, `density`, `ip`, `betan`, `q95`, `li` and `kappa`, respectively.

For each discharge, non-finite time-value pairs were removed, timestamps were ordered and duplicate or non-increasing samples were discarded. The admitted signals were restricted to their common temporal support and represented on an aligned 1,000-sample grid. Across the canonical cohort, retained intervals span approximately 4.08–6.02 s, corresponding to discharge-dependent spacings of approximately 4.08–6.03 ms. The canonical analysis therefore does not assume one universal physical time step across discharges.

Primitive signals were standardized separately within each discharge,

$$\tilde{x}_r^{(j)}(t) = \frac{x_r^{(j)}(t) - \mu_r^{(j)}}{\sigma_r^{(j)}},$$

and responses and retained derived coordinates were represented in their corresponding standardized coordinates before regression. Reported coefficients therefore characterize normalized observational coordinates and should not be interpreted directly as dimensional physical constants.

Three admissible trajectory realizations were examined for the descriptive study: aligned signals, univariate spline reconstructions and Rauch–Tung–Striebel reconstructions [17–19]. Derivatives and trajectory-relational coordinates were reconstructed consistently within each variant. Signal provenance, reconstruction parameters, common-support rules and coordinate definitions are given in Supplementary Note S7.

DIII-D contract-specific ontology, search and validation

The same 62-discharge scientific object was examined under two distinct contracts. Both instantiated finite relational ontologies from the admitted plasma quantities, but their information boundaries and qualified claims were different.

Descriptive contract. For q_{desc} , the response was standardized dW_{dia}/dt and the scientific objective was retrospective relational organization of the realized plasma record. Contemporaneous target-containing coordinates were therefore admissible. The finite ontology contained retained primitive quantities, temporal rates, algebraic ratios and first-order trajectory-relational coordinates. Sparse structural selection identified the common seven-coordinate support

$$C_{\text{desc}}^* = \left\{ D_{\kappa} W_{\text{dia}}, D_{\beta_N} W_{\text{dia}}, D_{\beta_N} \kappa, D_{\beta_N} \ell_i, \frac{q_{95}}{\kappa}, \dot{\beta}_N, \dot{\kappa} \right\}.$$

The support was shared across the 62 realizations, while coefficients were estimated separately within each discharge,

$$\tilde{y}^{(j)}(t) = c_0^{(j)} + \sum_{r=1}^7 c_r^{(j)} \tilde{\psi}_r^{(j)}(t) + \varepsilon^{(j)}(t).$$

Cohort-mean coefficients were used only to summarize the center of the realization-specific coefficient population. Reconstruction errors were computed using the corresponding discharge-specific fits.

Two selected coordinates, $D_{\kappa} W_{\text{dia}}$ and $D_{\beta_N} W_{\text{dia}}$, contain the contemporaneous response derivative. The fitted object was therefore classified as a target-containing implicit descriptive closure rather than an independent predictor. The alternative trajectory reconstructions test sensitivity of this relational organization to numerical realization, not prospective forecasting. Algebraic elimination, denominator conditioning, target-map conditioning, coefficient identifiability and reconstruction sensitivity are reported in Supplementary Note S7.

Reconstruction contract. For q_{rec} , the target was I_p . The information boundary was changed before coordinate construction: I_p and all target-dependent descendants were excluded from the explanatory coordinate ontology. The resulting finite explored frontier contained 141 relational coordinates.

Seven development discharges were used for structural selection. The search policy selected a ten-coordinate support, denoted REL10, which was frozen before the remaining 55 discharges were used for external structural evaluation. REL141 denotes the corresponding dense 141-coordinate relational reference. RAW10 denotes a size-matched ten-coordinate representation constructed without the expanded relational coordinate set.

For each external discharge, the transferred coordinate support was held fixed. Coefficients were calibrated using the first 80% of that discharge and reconstruction was evaluated on the protected final 20%. Thus the experiment tests transfer of coordinate structure rather than zero-shot transfer of one universal coefficient vector. The protected interval is a reconstruction holdout and not a prospective forecast.

Performance was summarized by pooled reconstruction error and by paired discharge-level comparison. REL10 achieved lower reconstruction error than RAW10 on 46 of 55 external discharges while retaining most of

the dense REL141 performance. Effective degrees of freedom were also retained as a complexity diagnostic. Complete candidate definitions, development-discharge membership, selected support, error calculations and structural-transfer diagnostics are given in Supplementary Note S7.

Reproducibility and qualified-result provenance

For every reported study, the reproducible computational object comprises more than the displayed relation. The archived analysis should contain the source record identifiers or generator specification; development and evaluation partitions; information boundary; mathematical-object interpretation; ontology and constructor versions; numerical coordinate realizations; search settings; explored candidate set; selected support; fitted coefficients; supported domain and masks; validation evidence; and known limitations.

Changes to preprocessing, derivative estimation, coordinate grammar, masks, search bounds or support after protected evaluation require a new discovery branch and a new protected evaluation procedure. This convention prevents adaptive representational choices from being hidden inside preprocessing. Formal provenance requirements and prospective ontology-refinement procedures are described in Supplementary Notes S1–S3 and S8.

Supplementary Information

S1 Formal structure of task-conditioned relational discovery

S1.1 Scientific records and questions

The scientific object in Eq. (??),

$$\mathcal{O} = (\mathcal{D}, \Omega_{\text{obs}}, \mathcal{S}, \mathcal{E}, \Pi, \mathcal{A}),$$

retains six logically distinct ingredients. The data component $\mathcal{D}$ contains stored values and masks; Ω_{obs} records the observed temporal, spatial, parameter or ensemble domain; $\mathcal{S}$ specifies sampling, discretization, ordering and boundary conventions; and $\mathcal{E}$ represents uncertainty, missingness and known error. Provenance Π contains the ordered lineage that produced the record. For an experiment it can include diagnostics, calibrations, synchronization and processing; for a simulation, the governing model, parameters, geometry, mesh, scheme, tolerances, software version and post-processing. Availability $\mathcal{A}$ records when, where and under which operational conditions each component can be used. It distinguishes stored information from information admissible to a retrospective, predictive or control task.

The question q is broader than a response variable. It can ask for an invariant, a parameter family, a reduced closure, a regime discriminator, a future forecast, an uncertainty-aware surrogate or a control action. The same record can support several questions, but their admissible information and validation need not agree. We therefore write

$$\mathcal{O}_q = \mathcal{I}_q(\mathcal{O})$$

for the information admissible to q .

S1.2 Discovery contracts, relational ontologies and admissible universes

A discovery contract is written

$$\mathcal{K}_q = (q, \mathcal{I}_q, \mathcal{P}_q, \mathcal{B}_q, \mathcal{H}_q, U_q, \mathcal{V}_q, \Omega_q).$$

The information operator first produces

$$\mathcal{O}_q = \mathcal{I}_q(\mathcal{O}),$$

after which the admissible record is assigned a mathematical interpretation

$$X_q = \text{Obj}_q(\mathcal{O}_q).$$

The scientific objective is encoded by $\mathcal{K}_q$, whereas X_q specifies the mathematical kind of object on which admissible constructors act.

Let Λ denote an extensible catalogue of typed coordinate and computational constructors and let $\mathfrak{T}$ denote an extensible catalogue of relation or model templates. The task-conditioned relational ontology is

$$\mathcal{G}_q = \Gamma(X_q, \mathcal{O}_q, \mathcal{K}_q; \Lambda, \mathfrak{T}).$$

The catalogues provide a vocabulary; $\mathcal{G}_q$ supplies the task-conditioned compositional grammar. In particular, the presence of a constructor in Λ does not imply that it is admissible for every object or task.

An individual coordinate c carries an extensible type

$$\tau(c) = (\text{kind}(c), \text{dim}(c), \text{unit}(c), \Omega_c, \mathcal{S}_c, \mathcal{E}_c, \mathcal{A}_c, \Pi_c, \dots).$$

A constructor $\lambda \in \Lambda$ has an input–output signature

$$\lambda : (\tau_1, \dots, \tau_r) \longrightarrow \tau_{\text{out}}$$

917 and an admissibility predicate

$$P_\lambda(X_q, \mathcal{O}_q, \mathcal{K}_q; c_1, \dots, c_r) = 1.$$

918 The predicate can encode mathematical domain restrictions, dimensional compatibility, information availability,
919 numerical conditioning, uncertainty, regularity or task-specific requirements.

920 The ontology induces the family of admissible coordinate systems

$$\mathfrak{C}_q = \text{Coord}(\mathcal{G}_q)$$

921 and, conditional on a coordinate system C ,

$$\mathfrak{R}_q(C) = \text{Rel}(\mathcal{G}_q, C).$$

922 The admissible coordinate–relation universe is therefore

$$\mathfrak{A}_q = \{(C, R) : C \in \mathfrak{C}_q, R \in \mathfrak{R}_q(C)\}.$$

923 This universe is epistemic rather than necessarily enumerable. A construction can be scientifically admissible
924 even if the computational search never reaches it. Conversely, a mathematically evaluable construction need
925 not belong to $\mathfrak{A}_q$ if it violates the semantic, information or uncertainty requirements of the contract.

926 **S1.3 Search, utility and qualified selection**

927 A search policy

$$\Sigma_q = \text{SearchPolicy}(\mathcal{G}_q, \mathcal{K}_q)$$

928 allocates finite computational effort within the admissible universe. Together with the search bounds $\mathcal{B}_q$, it
929 generates the explored frontier

$$\widehat{\mathfrak{A}}_q = \text{Explore}(\mathfrak{A}_q; \Sigma_q, \mathcal{B}_q) \subseteq \mathfrak{A}_q.$$

930 The distinction among

$$\mathfrak{A}_q, \quad \widehat{\mathfrak{A}}_q, \quad \mathcal{Q}_q^*$$

931 separates what is scientifically admissible, what was actually examined and what ultimately survived
932 qualification.

933 Candidate comparison is represented schematically by

$$(C^*, R^*) \in \text{Select}_q(\widehat{\mathfrak{A}}_q; U_q, \mathcal{V}_q).$$

934 The notation does not require U_q to be a scalar. A contract can use hard constraints, tolerance-defined
935 equivalence, lexicographic comparison, Pareto criteria or combinations of these. A scalar utility and ordinary
936 argmax are special cases.

937 The qualified result is written

$$\mathcal{Q}_q^* = (\mathcal{K}_q, C^*, R^*, \Omega^*, \Pi^*, \mathcal{A}^*, \mathcal{E}_{\text{val}}^*, \mathcal{L}^*).$$

938 The provenance component records, directly or by immutable reference, the ontology version, numerical
939 coordinate realizations, search policy, explored frontier and any external scientific knowledge used to prioritize
940 search.

941 This construction also bounds negative claims. Failure to discover a relation can establish failure within the
942 explored frontier; it does not establish that no admissible representation exists elsewhere in $\mathfrak{A}_q$. Similarly,
943 failure of one numerical realization does not necessarily invalidate the underlying semantic coordinate when
944 another admissible realization remains available.

S1.4 Admissible objects, coordinates and relations

Let

$$\mathcal{X}_q = \text{Obj}_q(\mathcal{O}_q)$$

be the observational object assigned to the record. The family of admissible coordinate systems is

$$\mathfrak{C}_q = \{C = \Phi_q(\mathcal{X}_q) : C \text{ is constructible from } \mathcal{O}_q \text{ and satisfies } \mathcal{K}_q\}.$$

Coordinates can include recorded quantities, derivatives, delays, integrals, transforms, reduced modes, dimensionless groups, invariants, geometric descriptors, event statistics, latent coordinates or relational derivatives. Constructibility is necessary but not sufficient: a centered smoother can be computed retrospectively but is unavailable to a strictly causal predictor.

For each C , $\mathfrak{R}_q(C)$ contains the permitted relations. A candidate can be an explicit map, implicit constraint, differential or differential-algebraic system, rational representation, stochastic coefficient family, integral relation, inequality, invariant, variational statement, nonparametric map or hybrid. The admissible search domain is

$$\mathfrak{A}_q(\mathcal{O}) = \{(C, R) : C \in \mathfrak{C}_q, R \in \mathfrak{R}_q(C), (C, R) \text{ satisfies } \mathcal{K}_q\}.$$

S1.5 Utility is task relative

The utility in Eq. (??) may combine several of the following:

- reconstruction or compression fidelity;
- structural or coefficient recovery in controlled data;
- stability across realizations, preprocessing and resolution;
- discrimination among branches, regimes or system states;
- generalization to unseen times, parameters, geometries or boundaries;
- predictive accuracy under causal information constraints;
- interpretability, parsimony and dimensional consistency; and
- decision-level or closed-loop performance.

These criteria are not interchangeable. A compact retrospective relation can be excellent for compression and inadmissible for forecasting. A model can predict accurately while failing to identify its coefficients or mechanism. The contract prevents the objective from being reinterpreted after seeing the result.

S1.6 Operational objectives, numerical admissibility and evidential escalation

The scientific utility of a representation is inseparable from the task for which it is constructed. Description, reconstruction, prediction and control are useful landmarks, but they are not an exhaustive taxonomy of scientific objectives, nor are they mutually exclusive. A discovery contract may instead seek an invariant, parameter family, reduced manifold, symmetry, regime classifier, uncertainty model, experimental discriminator or other qualified object. The four classes in Table S1 are useful because they illustrate how the same mathematical representation can acquire different evidential meanings as the information boundary and intended use are changed.

Task level is a property of the claim, not of the equation. The same fitted relation can participate in several contracts without supporting the same inference in each of them. Consider a general implicit representation

$$F(Y, X; \theta) = 0,$$

where Y denotes one or more quantities designated as targets under the current task and X denotes the remaining admissible coordinates. A small residual can be sufficient evidence for descriptive compatibility with the observed object. Reconstruction requires the stronger statement that the relation can be solved stably for Y using only the permitted X . Prediction further requires that X itself be constructible at the relevant information boundary. Decision or control additionally requires that use of the representation improve a declared objective after actions alter the trajectory of the system.

Consequently, admissibility is not determined by symbolic form alone. A coordinate containing the contemporaneous target can be valuable for description, compression, consistency testing or branch organization and be inadmissible for independent reconstruction of that same target. A centered smoother can be valid for retrospective reconstruction and invalid for forecasting. A highly accurate predictor can be unsuitable for control if small model errors are amplified by the induced feedback policy.

Target conditioning provides a numerical boundary between description and reconstruction.

The distinction becomes quantitative for a scalar target. Suppose elimination or coordinate substitution gives a target-affine relation

$$F(y, X) = A(X)y - B(X) = \epsilon.$$

Where $A \neq 0$, the implied reconstruction is

$$\hat{y} = \frac{B}{A},$$

and the implicit residual satisfies

$$y - \hat{y} = \frac{\epsilon}{A}.$$

Thus, in consistently standardized coordinates, the local inverse amplification associated with the target is proportional to

$$\Gamma_y = \frac{1}{|A|}.$$

An arbitrarily small implicit residual can therefore coexist with an unstable reconstruction when $|A|$ is small. This is precisely why residual fidelity and reconstructive utility cannot be identified with one another.

More generally, for

$$F(y, X) = 0,$$

the implicit-function derivative is

$$\frac{\partial y}{\partial X} = - \left(\frac{\partial F}{\partial y} \right)^{-1} \frac{\partial F}{\partial X},$$

where defined. The scalar quantity

$$A = \frac{\partial F}{\partial y}$$

therefore acts as a local numerical litmus test for whether an implicit description can safely be promoted to a reconstruction. In a task-conditioned search, this information can enter either as an admissibility constraint,

$$\inf_{\Omega_q} |A| \geq \tau_A,$$

or as a contribution to the utility that penalizes inverse amplification. A relation need not satisfy such a condition when the intended output is only a descriptive constraint; it generally must satisfy an appropriate analogue when the relation is expected to determine the target.

This distinction is particularly relevant to target-containing trajectory-relational coordinates. A phase-derivative representation can produce an exceptionally compact implicit description while its algebraically eliminated coefficient A approaches zero on parts of the trajectory. The representation can then be useful under a descriptive or compression contract even though it fails the numerical conditions required for stable target reconstruction. Conditioning is therefore part of the task definition rather than a post hoc property of a fitted equation.

Vector targets, fields and PDE representations. The same principle extends naturally beyond scalar reconstruction. Let

$$Y \in \mathbb{R}^m, \quad F(Y, X) \in \mathbb{R}^p, \quad p \geq m,$$

and define the target Jacobian

$$J_Y = D_Y F.$$

Local reconstruction requires J_Y to have full column rank. A natural inverse-amplification measure is

$$\Gamma_Y = \left\| J_Y^\dagger \right\|_2 = \frac{1}{\sigma_{\min}(J_Y)},$$

where $J_Y^\dagger$ is the Moore–Penrose inverse and $\sigma_{\min}$ is the smallest target-resolving singular value. The scalar A above is the one-dimensional instance of this condition.

This observation makes numerical well-posedness itself searchable. A reconstructive contract can restrict its admissible set to candidates satisfying

$$\sigma_{\min}(J_Y) \geq \tau_Y$$

over a declared portion of the validation domain, or can trade fidelity, complexity and inverse amplification through U_q . The search is then not merely for a relation with a small residual, but for a representation in which the desired inverse problem is numerically resolvable.

For fields and coupled systems, Y can be a discretized field, a block of state variables, a set of modal coefficients or a collection of derivative coordinates. A candidate PDE can be represented schematically as

$$\mathcal{F}(u, \partial_t u, \nabla u, \nabla^2 u, \dots) = 0.$$

If the contract asks only whether this relation compactly describes an observed field, descriptive residual and nonredundancy can be sufficient primary criteria. If the task instead requires evolution in time, one may designate $\partial_t u$, or a future field state, as the target and require the corresponding discrete target Jacobian to remain invertible and well-conditioned. In a continuum formulation the analogous object is the Fréchet derivative of the residual operator with respect to the designated target field; invertibility or suitable coercivity of the linearized operator plays the role of the finite-dimensional singular-value test.

Thus, task conditioning can alter a PDE search even when its symbolic grammar is unchanged. Two relations having nearly identical residuals can be ranked differently because only one defines a stable reconstruction or evolution problem. Weak forms, conservation constraints, integral relations, differential-algebraic systems and other representations can be treated in the same way by attaching numerical conditions appropriate to the role they are asked to perform.

Prediction introduces an information boundary and propagation problem. Reconstruction and prediction differ even when they use the same symbolic relation. Let $\mathcal{I}_t$ denote the information admissible at forecast time t . A predictive representation must have the form

$$\hat{Y}_{t+h} = \mathcal{P}(\mathcal{I}_t), \quad h > 0,$$

and every coordinate entering $\mathcal{P}$ must itself be constructible from $\mathcal{I}_t$. Target conditioning remains relevant, but it is no longer sufficient. Errors can accumulate under iteration, unstable modes can amplify small reconstruction errors, and a representation calibrated on one part of state space can encounter new regions during rollout.

A predictive contract can therefore supplement local target conditioning with tests of forecast-horizon stability, uncertainty calibration, temporal or realization-level holdout, sensitivity to admissible preprocessing and generalization under relevant changes of operating condition. For dynamical systems or discovered PDEs, stability of the inferred evolution operator can become part of the admissibility or utility calculation. SIR does not require one universal predictive criterion; it requires that the criterion match the forecast being claimed.

Decision and control move utility outside the fitted relation. For decision problems the quality of a representation is determined by what happens when it is used. If

$$a_t = \pi(\mathcal{I}_t)$$

denotes an action generated by a policy based on a selected representation, then a generic finite-horizon objective is

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{k=0}^H \ell(X_{t+k}, a_{t+k}) \right],$$

subject, where appropriate, to state, actuator, safety or resource constraints. Two representations with distinguishable reconstruction errors can be equivalent for this objective, while predictors with nearly identical forecast scores can induce substantially different actions.

The evidential burden therefore moves from model error to policy behavior. Relevant mathematical criteria can include controllability and observability, closed-loop stability, robustness margins, constraint satisfaction and sensitivity of the selected action to uncertainty in the fitted representation. In practical settings this can require simulation, off-policy evaluation, hardware-in-the-loop tests or controlled intervention before a decision-level claim is justified. Model-predictive control provides a familiar example in which optimization, hard constraints and stability of the receding-horizon policy are integral to the scientific and engineering utility of the representation [43].

Evidence escalates with the operational claim. The four illustrative classes therefore correspond to different evidential questions rather than simply different losses. A descriptive or compression claim requires, at minimum, mathematical and dimensional admissibility, nontriviality relative to coordinate definitions, fidelity on the declared object, provenance and stability sufficient for the intended description. A reconstructive claim additionally requires separation of target from admissible explanatory information, target solvability and conditioning, and evaluation on withheld target information. A predictive claim further requires a causal or otherwise prospectively valid information boundary, genuinely protected future or out-of-domain evaluation, and evidence that errors remain controlled over the forecast operation being claimed. A decision or control claim additionally requires action-level evaluation, constraint and robustness analysis, and evidence that the closed-loop or interventional use of the representation improves the declared utility.

Schematically,

$$\begin{array}{lcl} E_{\text{desc}} & \xrightarrow{\text{+ target exclusion and inverse stability}} & E_{\text{rec}} \\ & \xrightarrow{\text{+ prospective information and propagation tests}} & E_{\text{pred}} \\ & \xrightarrow{\text{+ action, robustness and closed-loop evidence}} & E_{\text{dec}}. \end{array}$$

These arrows indicate additional evidential obligations associated with the stronger operational claim; they do not assert a universal hierarchy of scientific value. A theorem establishing an invariant can be more important than a high-performing controller, and a compact equation of state need never become a forecasting model to be useful.

Classical representation-building as a searchable operation. Much of mathematical physics has advanced not only by solving equations in fixed variables, but by changing the representation in which a problem is posed. Variational formulations replace a trajectory equation by stationarity of a functional; Hamiltonian and Hamilton–Jacobi formulations reorganize dynamics around new state variables or characteristic functions [44]; perturbation and canonical-transformation methods seek coordinates in which dominant and weak effects separate; modal, spectral and asymptotic methods replace a field by coefficients or reduced orders; and Poincaré’s treatment of celestial mechanics provides a classical example in which changes of variables, perturbative constructions and invariant structure expose organization not apparent in the original equations [45].

SIR places this representational step inside the computational discovery contract. Formally, the admissible coordinate family

$$\mathcal{C}_q = \{\Phi(X_q) : \Phi \text{ is admissible under } K_q\}$$

can contain not only measured quantities but differentiation, integration, projection, nondimensionalization, weak-form operators, variational functionals, perturbative or symmetry-adapted transformations, spectral and modal representations, trajectory-relational coordinates, learned maps and compositions of such constructions. The associated relation family $\mathcal{R}_q(C)$ can likewise contain explicit equations, implicit constraints, PDEs, integral equations, stochastic parameter families, variational conditions, inequalities or nonparametric maps.

In this sense, classical mathematical representations can be absorbed as candidate coordinate and relation generators rather than treated as a single privileged language supplied before discovery. The computational advantage is that many such representational hypotheses can be generated, fitted, audited and compared under the same declared task. A reconstruction search can reject an otherwise excellent implicit relation because its target Jacobian is nearly singular; a predictive search can reject the same relation because one coordinate violates the forecast-time information boundary; and a control search can prefer a representation with slightly worse forecast error because it produces a more stable or robust policy.

This claim is architectural rather than exhaustive. No finite grammar spans all mathematically meaningful representations, and numerical search does not replace theorem proving, physical reasoning or the invention of new mathematical objects. Instead, SIR provides a mechanism by which established, learned and newly proposed representations can enter a common auditable search space, with the strength of the resulting claim determined by the task and evidence attached to it. The representational operations that historically entered science through analytic insight can therefore also serve as explicit computational hypotheses to be searched, combined and rejected rather than remaining fixed assumptions upstream of model discovery.

S1.7 Qualified outputs

A selected SIR result is reported as

$$\mathcal{Q}_q = (R, C, q, \mathcal{O}, \mathcal{V}_q, \mathcal{E}_{q,\text{val}}, \Omega_q, \Pi_q, \mathcal{A}_q, \mathcal{L}_q).$$

The tuple binds the symbolic or numerical relation to the coordinates in which it was discovered, the question it addresses, the evidence supporting it and the limitations that remain. A candidate absent from the output can have failed validation, violated admissibility or simply never entered the finite computational search. Consequently, the explored coordinate families, operator grammar, complexity bounds and optimizer must be archived alongside the selected model set.

S2 Observational objects, ambiguity and provenance

S2.1 From a source process to a coordinate system

Let $\mathcal{Y}$ denote one realization of a scientific process and $\mathcal{T}_\Pi$ the provenance-dependent operations that create the stored data:

$$\mathcal{D} = \mathcal{T}_\Pi(\mathcal{Y}).$$

The complete path is

$$\text{source process} \longrightarrow \text{realization} \longrightarrow \text{record} \longrightarrow \text{observational object} \longrightarrow \text{coordinate system}.$$

These stages are deliberately distinct. Experimental data are instrument mediated; simulations are model and discretization mediated; synthetic data are generator mediated; hybrid records can contain all three epistemic statuses.

If $\mathcal{T}_\Pi$ is not injective, distinct realizations can produce the same record. This induces

$$\mathcal{Y}_1 \sim_{\mathcal{O}} \mathcal{Y}_2 \iff \mathcal{T}_\Pi(\mathcal{Y}_1) = \mathcal{T}_\Pi(\mathcal{Y}_2).$$

Relational discovery can characterize structure preserved by the resulting equivalence class but cannot select a unique source realization without additional information. A subsequent coordinate map Φ_q can remove further distinctions or, through derivatives, delays or lifting, refine some that were collapsed in a value-only projection [20, 21].

S2.2 Object classes and natural operators

Table S2: Illustrative observational objects and coordinate constructions. The table guides admissibility; it is not an exhaustive or exclusive taxonomy.

Object	Examples	Natural coordinate constructions
Trajectory	Diagnostic signals, state histories, reduced simulation paths	Derivatives, delays, integrals, tangents, curvature, recurrence, trajectory-relational coordinates
Spatial or spatiotemporal field	Temperature, density, displacement, flux, finite-element solution	Gradients, divergence, curl, Laplacians, fluxes, weak forms, modal coefficients, integral functionals
Profile family	Radial plasma profiles, boundary-layer sections	Shape parameters, basis coefficients, extrema, widths, aligned gradients, profile-to-profile distances
Point or event process	ELMs, failures, arrivals, photon or neutron counts	Counts, intensities, hazards, waiting times, marks, event-conditioned histories
Graph or network	Diagnostic coupling, interaction or transport networks	Graph gradients, Laplacians, centralities, motifs, spectral embeddings
Ensemble or distribution	Parameterized simulations, repeated experiments, stochastic realizations	Moments, quantiles, empirical measures, coefficient distributions, transport maps
Geometric object	Boundaries, flux surfaces, meshes, manifolds	Curvature, topology, signed distances, shape modes, symmetry-adapted coordinates
Hybrid coupled system	Multimodal experiments and experiment-simulation records	Cross-object reductions, synchronized coordinates, coupled constraints, provenance-aware relational operators

S2.3 The observational-object audit

The audit asks:

1. What was directly recorded, and what was reconstructed or simulated?
2. What are the natural independent domains and ensemble indices?
3. Which distinctions can sampling, averaging or projection have removed?
4. Which operators are mathematically meaningful for the resulting object?
5. Which coordinates are available under the intended task?

6. Which upstream sources or reconstructions are shared among variables?
7. Which uncertainty, mask and boundary conventions propagate to derived coordinates?

This audit prevents the convenience of an array layout from silently deciding the mathematics. For example, a profile stack can be differentiated in time, space or parameter; each operator answers a different question and requires a different boundary and sampling interpretation.

S2.4 Provenance and information availability

For coordinate u_i , write

$$\Pi(u_i) = (s_i, \mathcal{T}_i, \boldsymbol{\lambda}_i, \mathcal{D}_i),$$

where s_i is its source, $\mathcal{T}_i$ the ordered transformations, $\boldsymbol{\lambda}_i$ their settings and $\mathcal{D}_i$ dependencies on other quantities or datasets. Two identically named coordinates are not the same computational object when constructed with different calibrations, smoothing scales, normalizations, meshes or solvers.

For an online prediction at time t , a causal coordinate must satisfy

$$u(t) = \phi(\mathcal{O}_q|_{(-\infty, t]}).$$

Centered differences, full-trajectory smoothing, global normalization fitted with held-out observations and post-event labels violate this condition. They can remain admissible for retrospective analysis. Equivalent restrictions hold for surrogates intended to run before a high-fidelity simulation: inputs cannot be constructed from the completed solution they are meant to replace.

S3 Computational construction and search

S3.1 Recursive coordinate grammar

Starting from

$$\mathcal{B}_\alpha^{(0)} = \{1, u_{\alpha,1}, \dots, u_{\alpha,N_\alpha}\},$$

let $\mathfrak{G}_q^{(1)}$, $\mathfrak{G}_q^{(2)}$ and $\mathfrak{G}_q^{(D)}$ contain admissible unary, binary and object-dependent operators. A recursive library is

$$\begin{aligned} \mathcal{B}_\alpha^{(\ell+1)} = & \mathcal{B}_\alpha^{(\ell)} \cup \{g(\psi) : g \in \mathfrak{G}_q^{(1)}, \psi \in \mathcal{B}_\alpha^{(\ell)}\} \\ & \cup \{h(\psi_i, \psi_j) : h \in \mathfrak{G}_q^{(2)}, \psi_i, \psi_j \in \mathcal{B}_\alpha^{(\ell)}\} \\ & \cup \{D\psi : D \in \mathfrak{G}_q^{(D)}, \psi \in \mathcal{B}_\alpha^{(\ell)}\}, \end{aligned}$$

with admissibility evaluated at each step. Candidate complexity can be assigned recursively:

$$\kappa(u_i) = 1, \quad \kappa(g(\psi)) = \kappa(\psi) + c(g), \quad \kappa(h(\psi_i, \psi_j)) = \kappa(\psi_i) + \kappa(\psi_j) + c(h).$$

The finite library retains only terms with $\kappa(\psi) \leq K$ and admissibility indicator $A_q(\psi) = 1$.

Dimensional signatures can be propagated through the grammar. Addition requires matching signatures, multiplication adds exponent vectors, division subtracts them and logarithmic or exponential transforms generally require dimensionless arguments. These restrictions both improve scientific interpretation and reduce search.

S3.2 Search policies, proposal priority and explored-frontier provenance

Admissibility does not require that every reachable construction receive equal computational effort. Let ξ denote an admissible coordinate construction, relation template or partial compositional program. A search

policy may associate with ξ a task-conditioned proposal priority

$$\pi_q(\xi \mid X_q, \mathcal{G}_q, \mathcal{H}_q).$$

The quantity π_q is schematic: depending on the implementation it can be a scalar score, partial ordering, proposal distribution, acquisition function or learned expansion policy.

Priority and utility have different evidential roles. The proposal priority controls search allocation,

$$\pi_q : \text{where to search first,}$$

whereas the contract utility and validation determine

$$(U_q, \mathcal{V}_q) : \text{which examined candidates are scientifically supported.}$$

A high-priority construction therefore carries no additional validation weight merely because it was suggested by established theory or frequently appears in the scientific literature.

The external knowledge object $\mathcal{H}_q$ can include peer-reviewed literature, curated databases, accepted theory, previous qualified discoveries, laboratory records or other task-admissible scientific resources. If used, its source, version or retrieval date, extraction procedure and transformation into search priority are part of provenance. This allows scientific precedent to accelerate exploration without making precedent a criterion of truth.

For a recursively generated ontology, one possible priority-limited expansion is

$$\mathcal{N}_q^{(d+1)} = \text{Expand}_{\Sigma_q} \left(\mathcal{N}_q^{(d)}; \Lambda, \mathcal{P}_q, \mathcal{B}_q \right),$$

where Σ_q can allocate expansion budget preferentially to high-priority candidates while retaining an exploration channel for admissible low-priority regions. The resulting explored frontier

$$\hat{\mathfrak{A}}_q$$

must retain enough provenance to distinguish candidates that were evaluated, candidates rejected by an admissibility predicate and candidates that were never reached.

If priority is used only to order an otherwise exhaustive finite search, it does not change the final candidate universe. If priority is used to prune, truncate or allocate unequal computational budget, the associated rule becomes part of $\mathcal{B}_q$ and must be reported with any negative discovery claim.

The studies in the present Article use finite, transparent search policies and do not employ corpus-derived or learned π_q . This distinction is deliberate: the experiments test the scientific architecture while avoiding confounding by an opaque proposal mechanism.

S3.3 Evaluated relation matrices

The numerical evaluation of term $\psi_{\alpha,k}$ on the admissible record is a column $\mathbf{v}_{\alpha,k}$. Stacking columns gives

$$\Psi_\alpha = [\mathbf{v}_{\alpha,1} \quad \cdots \quad \mathbf{v}_{\alpha,K_\alpha}].$$

Rows can be time samples, spatial points, quadrature functionals, events or realization-level reductions. Multi-realization matrices can be pooled when coefficients are shared or retained as blocks when only structural support is shared.

Two terms can be annotated as numerically near-equivalent if

$$\min_{c \neq 0} \frac{\|\mathbf{v}_i - c\mathbf{v}_j\|}{\|\mathbf{v}_i\|} \leq \varepsilon_{\text{eq}}.$$

Rank-revealing decompositions and condition numbers diagnose broader dependencies. Removing a column improves conditioning but can erase a scientifically meaningful alternative; the decision is therefore part of the contract.

S3.4 Explicit, implicit and rational estimation

For an explicit system with response matrix $\mathbf{Y}$, unregularized least squares gives

$$\hat{\mathbf{A}} = \mathbf{\Psi}^+ \mathbf{Y},$$

with $(\cdot)^+$ the Moore–Penrose pseudoinverse. Sparse penalties or sequential thresholding can replace the unregularized solution [3, 22, 23].

For an implicit homogeneous relation,

$$\mathbf{\Psi} \mathbf{a} \approx \mathbf{0},$$

the coefficient vector can be obtained from a constrained singular-vector, null-space or sparse implicit regression problem. A normalization such as $\|\mathbf{a}\|_2 = 1$, or fixing one coefficient on each chart, excludes the zero vector but does not change the relational zero set.

For rational regression,

$$y = \frac{\sum_{i=1}^m a_i P_i}{\sum_{j=1}^n b_j Q_j},$$

an exact relation can be represented linearly in the coefficients:

$$\sum_{i=1}^m a_i P_i - y \sum_{j=1}^n b_j Q_j = 0.$$

Approximate fitting may instead minimize a task loss on the evaluated quotient, subject to coefficient normalization and a denominator-domain constraint. Cross-multiplication can change behavior at denominator zeros; the supported domain is therefore part of the relation.

S3.5 Combinatorial control and iterative construction

If m primitive quantities are combined with n unary forms up to interaction order n_ℓ , even a simplified monomial count grows as

$$C(n, m, n_\ell) = \sum_{i=0}^{n_\ell} \binom{1 + nm}{i}.$$

Hard constraints can fix denominator form, derivative order, interaction order or required physical terms. Greedy forward selection reduces the number of conditional regressions substantially, but can miss combinations whose terms are useful only jointly. Randomized starts, backward pruning and stability selection partially probe this dependence.

Search can also be iterative:

$$\text{coordinates} \rightarrow \text{relations} \rightarrow \text{estimation} \rightarrow \text{validation} \rightarrow \text{revised coordinates}.$$

A retained composite term can become a coordinate in the next stage. No information from a protected evaluation set may be used to revise the search without redefining that set as development data.

S3.6 Coefficient families

When one structure is shared across realizations but coefficients vary, the output is

$$\{\boldsymbol{\theta}^{(1)}, \dots, \boldsymbol{\theta}^{(J)}\}.$$

Empirical distributions, kernel density estimates, hierarchical models or covariate-conditioned coefficient manifolds can summarize this collection. The distinction among within-realization estimation error, between-realization variability and structural uncertainty should be preserved. The oscillator example isolates this setting with a known one-dimensional parameter distribution.

S4 Mathematics of trajectory-relational coordinates

S4.1 Classical definition and local interpretation

Let $\gamma : I \rightarrow \mathcal{M}$ be C^1 , and let $f, g : \mathcal{M} \rightarrow \mathbb{R}$ be C^1 near $\gamma(I)$. Define

$$f_\gamma = f \circ \gamma, \quad g_\gamma = g \circ \gamma.$$

At t_0 with $g'_\gamma(t_0) \neq 0$, the inverse function theorem gives a neighborhood U on which $g_\gamma|_U$ is invertible. The derivative of f with respect to g along γ is then

$$D_g^\gamma f(t_0) = \frac{d}{dz} [f_\gamma \circ (g_\gamma|_U)^{-1}](z) \Big|_{z=g_\gamma(t_0)} = \frac{f'_\gamma(t_0)}{g'_\gamma(t_0)}.$$

Thus, the word derivative has its usual local meaning even when f is not a globally single-valued function of g .

In tangent notation,

$$D_g^\gamma f(t) = \frac{(df)_{\gamma(t)}[\dot{\gamma}(t)]}{(dg)_{\gamma(t)}[\dot{\gamma}(t)]}.$$

Both differentials are evaluated on the same tangent vector. Let $\varphi : J \rightarrow I$ be a regular C^1 reparameterization, with $\varphi'(\tau) \neq 0$, and define $\tilde{\gamma} = \gamma \circ \varphi$. Then

$$\tilde{D}_g^\gamma f(\tau) = \frac{f'_\gamma(\varphi(\tau))\varphi'(\tau)}{g'_\gamma(\varphi(\tau))\varphi'(\tau)} = D_g^\gamma f(\varphi(\tau)).$$

The coordinate is therefore invariant to regular changes in traversal rate, including orientation reversal. This does not imply invariance under arbitrary transformations or rescalings of the observables f and g .

S4.2 Weaker regularity and finite observations

If f_γ and g_γ are absolutely continuous, their derivatives exist almost everywhere, and the quotient can be defined almost everywhere on the set where $g'_\gamma \neq 0$. Weak or distributional derivatives alone do not generally define a quotient of distributions. A claimed weak phase derivative must therefore specify the relevant function spaces, representatives and multiplication or regularization rules.

For finite observations, two distinct constructions are useful. The discrete secant slope is

$$\widehat{D}_g f_{k+1/2}^{\text{sec}} = \frac{f_{k+1} - f_k}{g_{k+1} - g_k},$$

whereas a numerical estimate of the classical phase derivative is

$$\widehat{D}_g f(t_k) = \frac{\widehat{f'_\gamma}(t_k)}{\widehat{g'_\gamma}(t_k)}.$$

The first is a slope between observed points in the (g, f) plane; the second is a ratio of estimated local derivatives. They coincide only under additional limiting or interpolation assumptions.

Under sufficient smoothness, consistent order- p derivative estimates and a lower bound

$$|g'_\gamma(t)| \geq c_g > 0,$$

the ratio estimator inherits order- p convergence on the corresponding domain. Its error bound deteriorates as c_g approaches zero. The sampling grid, interpolation basis, smoothing scale, derivative stencil, boundary convention and denominator threshold therefore form part of the coordinate's provenance.

1253 S4.3 Higher-order and mixed constructions

1254 Define relational differentiation along a regular g -chart by

$$\mathcal{D}_g q = \frac{1}{g'_\gamma} \frac{dq}{dt}.$$

1255 The second iterated relational derivative is

$$\mathcal{D}_g^2 f = \frac{1}{g'_\gamma} \frac{d}{dt} \left(\frac{f'_\gamma}{g'_\gamma} \right) = \frac{f''_\gamma g'_\gamma - f'_\gamma g''_\gamma}{(g'_\gamma)^3}.$$

1256 This is the ordinary second derivative of f_γ with respect to the local coordinate g_γ .

1257 It differs from the same-order temporal-derivative ratio

$$R_g^{(2)} f := \frac{f''_\gamma}{g''_\gamma}.$$

1258 For orders $k \geq 2$, the quantity

$$R_g^{(k)} f = \frac{f_\gamma^{(k)}}{g_\gamma^{(k)}}$$

1259 is generally parameterization dependent and should not be identified with the k th derivative of f with respect
1260 to g .

1261 Sequential mixed constructions can be defined by

$$\mathcal{D}_h \mathcal{D}_g f = \frac{1}{h'_\gamma} \frac{d}{dt} \left(\frac{f'_\gamma}{g'_\gamma} \right).$$

1262 In general,

$$\mathcal{D}_h \mathcal{D}_g f \neq \mathcal{D}_g \mathcal{D}_h f.$$

1263 A one-dimensional trajectory supplies only one tangent direction and does not identify independent partial
1264 derivatives of a hypothetical multivariate function $F(g, h)$. Products such as

$$(D_g f)(D_h f)$$

1265 are composite relational coordinates, not mixed derivatives.

1266 S4.4 Lifting and topology of the represented image

1267 Let

$$P_{f,g}(t) = (f_\gamma(t), g_\gamma(t))$$

1268 be a planar observational projection. If

$$P_{f,g}(t_a) = P_{f,g}(t_b)$$

1269 but

$$D_g^\gamma f(t_a) \neq D_g^\gamma f(t_b),$$

1270 then the lifted map

$$\tilde{P}_{f,g}(t) = (f_\gamma(t), g_\gamma(t), D_g^\gamma f(t))$$

1271 separates those preimages. In terms of the equivalence classes induced by the coordinate maps,

$$[t]_{f,g,D_g f} \subseteq [t]_{f,g}.$$

1272 The inclusion can be strict at a projected self-intersection.

First-order lifting does not necessarily separate every degenerate pair. If two passages have the same tangent ratio, their first-order phase coordinates also coincide. Overlapping arcs, tangential self-intersections and branches having identical first-order jets may require delays, higher-order coordinates, additional observables or a different coordinate chart. If the available observations agree under all admissible constructions, the states are observationally non-identifiable and should not be artificially distinguished.

If a continuous lifted map is injective on compact I , it is a homeomorphism from I to its image. If it is also an immersion, its image is an embedded curve. A noninjective planar projection can instead have a self-intersecting image. Relational lifting can therefore change the topology of the *represented image* by removing projection-induced identifications. It does not change the topology of the source trajectory.

S4.5 Singularities, regularization and local charts

At $g'_\gamma = 0$, the classical coordinate is undefined unless a separate limit argument supplies a continuous extension. A primary numerical policy is masking:

$$\widehat{\Omega}_{f|g}^{(\tau)} = \left\{ t_k : \left| \widehat{g}'_\gamma(t_k) \right| > \tau_g \right\}.$$

The masked fraction and sensitivity to τ_g should be reported because turning points and small-denominator regions can themselves be scientifically meaningful.

Alternative finite coordinates include the denominator-shifted quotient

$$P_\delta^{\text{den}} = \frac{f'_\gamma}{g'_\gamma + \delta_g},$$

the affine-shifted quotient

$$P_{\delta_f, \delta_g}^{\text{aff}} = \frac{f'_\gamma + \delta_f}{g'_\gamma + \delta_g},$$

and the bounded quotient

$$P_\lambda^{\text{bd}} = \frac{f'_\gamma g'_\gamma}{(g'_\gamma)^2 + \lambda^2}.$$

These constructions can be useful for visualization, conditioning or regression, but they are not identical to the classical phase derivative. Additive shifts generally break reparameterization invariance and must have units compatible with their respective derivative terms. A data-dependent shift also depends on the observation window, signal scaling and derivative estimator.

A piecewise alternative preserves the classical coordinate on its regular domain and uses another coordinate Q elsewhere:

$$P(t) = \begin{cases} f'_\gamma(t)/g'_\gamma(t), & |g'_\gamma(t)| > \tau_g, \\ Q(t), & |g'_\gamma(t)| \leq \tau_g. \end{cases}$$

This construction suggests an atlas of local trajectory coordinates rather than one globally privileged quotient. Chart domains, transition rules, branch choices and continuity conventions then become part of the qualified representation.

S4.6 Sensitivity-centered trajectory-relational coordinates

The classical phase derivative

$$D_g^\gamma f = \frac{\dot{f}}{\dot{g}}$$

is defined only on the regular domain of the reference derivative. For global visualization or conditioning-sensitive representation, it can be useful to construct a related finite coordinate that retains the distinction between the target and reference rates without claiming the invariance properties of the classical quotient.

1304 Let n denote the numerator rate and d the reference rate, and let $S_n, S_d > 0$ be declared scale factors. Define

$$u_n = \frac{n}{S_n}, \quad u_d = \frac{d}{S_d}.$$

1305 A nonnegative reference shift is chosen from development support,

$$s_0 = \max\left(0, -\min_{\text{dev}} u_d\right), \quad s_{\text{eff}} = s_0 + \kappa,$$

1306 where $\kappa > 0$ supplies a fixed positive margin. Let

$$w = u_d + s_{\text{eff}}$$

1307 and define the bounded reference response

$$g(w) = \frac{w}{w^2 + \rho^2}, \quad \rho > 0.$$

1308 The reference-shifted coordinate is

$$C_{\text{RS}} = (u_n + s_{\text{eff}}) g(w).$$

1309 Writing

$$\bar{g} = \text{mean}_{\text{dev}} g(w),$$

1310 the sensitivity-centered coordinate is

$$C_{\text{SC}} = C_{\text{RS}} - \bar{g} u_n = u_n (g - \bar{g}) + s_{\text{eff}} g.$$

1311 The centering has a direct sensitivity interpretation:

$$\frac{\partial C_{\text{SC}}}{\partial u_n} = g - \bar{g}.$$

1312 Hence the direct numerator sensitivity vanishes on the conditioning locus

$$g = \bar{g}.$$

1313 At a turning event of the target, $n = 0$ and therefore $u_n = 0$, so the exact turning manifold is

$$C_{\text{SC}} = s_{\text{eff}} g(w).$$

1314 Thus a turning condition that can be strongly folded in another coordinate projection becomes organized
1315 around an explicitly characterized one-dimensional manifold.

1316 For Fig. 2, $n = \dot{x}_1$ and $d = \dot{x}_2$. The displayed construction uses the fixed conditioning parameters $\rho = 0.10$
1317 and $\kappa = 1$, with scaling, shifts and centering computed only from the designated construction domain. The
1318 coordinate is used as a representational example; it is not asserted to be a reparameterization-invariant
1319 replacement for the classical phase derivative.

1320 More generally, this construction illustrates that trajectory-relational coordinates can perform a geometric
1321 simplification rather than merely append a new feature. A complicated branch organization in one coordinate
1322 system can be represented as deviations from a low-complexity relational manifold in another. The scientific
1323 value of that simplification remains contract dependent.

1324 S4.7 Controlled multi-crossing trajectory

1325 The trajectory in Fig. ?? was constructed to contain multiple analytically controlled degeneracies while
1326 avoiding the appearance of a reflection-symmetric mathematical toy. For $t \in [0, 2\pi)$, define

$$\gamma(t) = F(r(t) \cos \phi(t), r(t) \sin \phi(t)), \quad r(t) = r_e(t) + r_o(t),$$

where

$$r_e(t) = 1.92 + 0.43 \cos(2t) + 0.26 \sin(2t) + 0.14 \cos(4t) \\ - 0.16 \sin(4t) + 0.07 \cos(6t),$$

$$r_o(t) = 0.60 \cos(3t) + 0.08 \sin t - 0.04 \cos t \\ - 0.07 \cos(5t) + 0.08 \sin(5t),$$

and

$$\phi(t) = 2t + 0.30 \sin(2t) - 0.10 \cos(2t) + 0.08 \sin(4t).$$

The nonlinear deformation F is

$$F(u, v) = (X, v - 0.28X + 0.05X^2 - 0.01X^3),$$

with

$$X = u + 0.16v + 0.06v^2 - 0.01v^3.$$

The radial and angular components satisfy

$$r_e(t + \pi) = r_e(t), \quad r_o(t + \pi) = -r_o(t),$$

and

$$\phi(t + \pi) = \phi(t) + 2\pi.$$

Numerical verification over the complete parameter domain gives

$$\min_t r(t) \approx 0.641 > 0, \quad \min_t \phi'(t) \approx 1.359 > 0, \quad \min_t \|\dot{\gamma}(t)\| \approx 1.010 > 0.$$

The undeformed polar trajectory therefore has positive radius, a strictly increasing twice-wound angle and nonzero velocity.

The deformation F is a composition of two triangular shears. Writing

$$X = u + h(v), \quad Y = v + k(X),$$

its inverse is obtained sequentially from

$$v = Y - k(X), \quad u = X - h(v).$$

Thus F is globally invertible and neither creates nor removes self-intersections.

Because ϕ is strictly increasing and advances by 2π over every parameter interval of length π , the only distinct parameter values having the same polar direction are paired as t and $t + \pi$. The paired points coincide precisely when their radii agree:

$$r(t) = r(t + \pi).$$

Using the periodicity properties above, this condition becomes

$$r_e(t) + r_o(t) = r_e(t) - r_o(t),$$

and hence

$$r_o(t) = 0.$$

Root isolation of the fixed trigonometric polynomial r_o produces exactly three simple zeros on $[0, \pi)$:

$$t_A \approx 0.578216, \quad t_B \approx 1.499442, \quad t_C \approx 2.658981.$$

Their paired preimages are $t_i + \pi$. The nonzero angles between the two tangent lines establish that all three coincidences are transverse double points.

Let

$$p(t) = D_{x_2}^\gamma x_1(t) = \frac{\dot{x}_1(t)}{\dot{x}_2(t)}$$

denote the classical relational coordinate at the crossings. For the global visualization, define

$$p_s(t) = D_{x_2}^s x_1(t) = \frac{\dot{x}_1(t) + s}{\dot{x}_2(t) + s},$$

where

$$s = 1 + \left\| \min_{\substack{t \in [0, 2\pi] \\ j \in \{1, 2\}}} \dot{x}_j(t) \right\| = 8.$$

This choice makes the shifted numerator and denominator strictly positive over the displayed dimensionless trajectory.

The crossing-level diagnostics are summarized in Table S3.

At every marked crossing,

$$\gamma(t_i) = \gamma(t_i + \pi)$$

but

$$p(t_i) \neq p(t_i + \pi), \quad p_s(t_i) \neq p_s(t_i + \pi).$$

The two lifts therefore refine each two-element equivalence class of the planar representation into distinct lifted states.

For a regular time reparameterization $t = \varphi(\tau)$, the classical coordinate transforms as

$$\frac{\frac{d}{d\tau} x_1(\varphi(\tau))}{\frac{d}{d\tau} x_2(\varphi(\tau))} = \frac{\dot{x}_1(\varphi(\tau))\varphi'(\tau)}{\dot{x}_2(\varphi(\tau))\varphi'(\tau)} = p(\varphi(\tau)).$$

The shifted coordinate instead becomes

$$\tilde{p}_s(\tau) = \frac{\dot{x}_1(\varphi(\tau))\varphi'(\tau) + s}{\dot{x}_2(\varphi(\tau))\varphi'(\tau) + s},$$

which is not generally equal to $p_s(\varphi(\tau))$. The shifted lift in Fig. ?? should therefore be interpreted as a globally finite visualization and as an example of a distinct admissible coordinate, not as a reparameterization-invariant extension of the classical phase derivative.

S4.8 Target-containing coordinates and induced implicit models

Let

$$u = \dot{x}, \quad v = \dot{y}, \quad D_x y = \frac{v}{u}.$$

The apparently linear model

$$v = c_0 + c_1 u + c_2 D_x y$$

is equivalent, where defined, to

$$(u - c_2)v - c_0 u - c_1 u^2 = 0,$$

or, when $u \neq c_2$, to

$$v = \frac{c_1 u^2 + c_0 u}{u - c_2}.$$

The derived coordinate contains v , so the fit is a low-order rational implicit parameterization rather than independent predictive evidence about v . It is not automatically tautological.

1370 In the lifted space (u, v, p) , the coordinate definition is

$$H(u, v, p) = up - v = 0,$$

1371 whereas the fitted relation is

$$G(u, v, p) = v - c_0 - c_1u - c_2p = 0.$$

1372 At regular points, G supplies additional local constraint content when

$$\text{rank } D(H, G) > \text{rank } DH.$$

1373 If the rank does not increase, or if the additional singular value is numerically unresolved, the fitted relation
 1374 is algebraically redundant or nearly degenerate. If G follows identically from H , it is tautological. These
 1375 structural conclusions do not follow merely from the appearance of v on both sides of the original displayed
 1376 equation.

1377 For the affine-shifted coordinate

$$P_\delta = \frac{v + \delta}{u + \delta},$$

1378 the model

$$v = c_0 + c_1u + c_2P_\delta$$

1379 rearranges to

$$v = \frac{(c_0 + c_1u)(u + \delta) + c_2\delta}{u + \delta - c_2}.$$

1380 The shift changes the conditioning and the induced rational family but does not remove target dependence.
 1381 The tuple (u, v, P_δ) lies on a two-dimensional algebraic surface in a three-dimensional ambient space.

1382 Target containment is task relative. The coordinate

$$(f, g, D_g^\gamma f)$$

1383 can separate value-space branches because $D_g^\gamma f$ is not generally determined by the values (f, g) . A phase
 1384 derivative constructed only from explanatory signals can also be used for a future target without algebraic
 1385 leakage, provided its numerical construction respects the intended information boundary.

1386 Usefulness requires three additional checks. First, local solution for the target v requires

$$\frac{\partial G_{\text{elim}}}{\partial v} \neq 0,$$

1387 or a full-rank target Jacobian in the vector case. Second, the inverse problem must remain conditioned away
 1388 from $u = c_2$ and other denominator zeros. Third, every coordinate must be available under the intended task.
 1389 A target-containing coordinate can therefore be valuable for retrospective compression, constraint discovery,
 1390 denoising, consistency testing or branch organization while remaining inadmissible as an independent predictor
 1391 of the same contemporaneous target.

1392 S4.9 Relation to established mathematics

1393 Ratios such as $\dot{y}/\dot{x}$ occur in classical phase-plane elimination [24, 25]. The local inverse-function interpretation
 1394 is standard differential calculus, and the tangent ratio has a differential-geometric expression [26]. Lie-
 1395 derivative ratios connect to nonlinear observability [27], while differential algebra, differential invariants,
 1396 moving frames and jet spaces contain closely related constructions [11, 12, 28, 29].

1397 The novelty claimed here is therefore not a new elementary derivative operator. It is the admission of such
 1398 quantities, together with other object-appropriate constructions, as task-dependent coordinates inside an
 1399 auditable joint coordinate-relation discovery framework.

S5 Representational equivalence, identifiability and claim calibration

S5.1 Sources of nonuniqueness

Representational nonuniqueness can enter at several distinct levels:

1. *source nonuniqueness*: distinct processes or realizations produce the same record;
2. *object nonuniqueness*: one record admits several mathematical interpretations;
3. *coordinate nonuniqueness*: one object admits several useful coordinate systems;
4. *relational nonuniqueness*: several relations agree on a fixed coordinate record;
5. *algebraic nonuniqueness*: different expressions define the same relational set on a domain;
6. *construction-induced nonuniqueness*: derived coordinates create exact or near-exact dependencies;
7. *finite-data nonuniqueness*: candidates differ outside the sampled domain but not at available resolution; and
8. *computational nonuniqueness*: preprocessing, search truncation or optimization produces different evaluated candidates.

These cases require different remedies. More accurate optimization cannot resolve a noninjective observation map; more realizations can resolve a finite-data coincidence but not necessarily an algebraic equivalence; and an intervention can distinguish mechanisms that passive observation cannot.

S5.2 Relational, coordinate and observational equivalence

For relation R on coordinate domain Ω_C , define

$$Z_{\Omega_C}(R) = \{\mathbf{c} \in \Omega_C : R(\mathbf{c}) = 0\}.$$

Relations R_1 and R_2 are relationally equivalent on Ω_C when their zero sets agree. Multiplication by a nowhere-zero factor preserves this set; multiplication by a factor that vanishes can add solutions. Cross-multiplication and division must therefore retain their excluded domains.

Coordinate systems C_α and $C_\beta = T(C_\alpha)$ are equivalent on a domain when T and T^{-1} are both admissible there. A relation transfers as $R_\beta = R_\alpha \circ T^{-1}$. Local branch equivalence does not imply global equivalence: a trajectory lift can deliberately remove an identification introduced by a noninjective projection.

Two qualified representations can be observationally equivalent without being mathematically equivalent. If $\text{Eval}_\mathcal{O}(\mathcal{Q})$ denotes the task-relevant residual, prediction, reconstruction, classification or decision, then

$$d_\mathcal{O}(\text{Eval}_\mathcal{O}(\mathcal{Q}_1), \text{Eval}_\mathcal{O}(\mathcal{Q}_2)) \leq \varepsilon_\mathcal{O}$$

defines equivalence at the declared resolution. Agreement on finitely many points is weaker than agreement on held-out realizations or across plausible coordinate reconstructions.

Task equivalence,

$$\mathcal{Q}_1 \sim_q \mathcal{Q}_2,$$

requires only that the candidates preserve distinctions material to q and have indistinguishable validated utility. Algebraically different models can be task equivalent for control, whereas algebraically equivalent forms can fail to be task equivalent if one is numerically unstable or unavailable online.

S5.3 Contract-relative identifiability

Parameter identifiability assumes a fixed coordinate system and structure. For homogeneous implicit relations, the coefficient vector is identifiable only up to nonzero scale unless a normalization is imposed. Structural identifiability asks whether the term support and operator architecture are unique up to a declared structural equivalence. Coordinate identifiability asks whether the contract privileges one representation. Mechanistic identifiability is stronger: it requires that competing generators be distinguishable through the available observations or interventions [30, 31].

Let

$$\mathfrak{Q}_q(\mathcal{O}) = \{\mathcal{Q}_q : \mathcal{Q}_q \text{ satisfies } \mathcal{K}_q \text{ and is supported by } \mathcal{O}\}.$$

The problem is identifiable at the resolution of q when

$$\mathfrak{Q}_q(\mathcal{O})/\sim_q = \{[\mathcal{Q}_q^*]_q\}.$$

Otherwise the quotient is a contract-relative ambiguity set. This definition does not demand one unique symbolic string; it demands that remaining representations be equivalent for the distinctions the question was designed to resolve.

New observations can be designed to separate the ambiguity set. If $\mathcal{D}_{\text{new}}$ is a candidate experiment or simulation, one can seek

$$\mathcal{D}_{\text{new}}^* \in \arg \max_{\mathcal{D}_{\text{new}}} \text{Sep}(\mathfrak{Q}_q/\sim_q; \mathcal{D}_{\text{new}}),$$

where Sep measures expected discrimination among the currently supported alternatives.

S5.4 Evidential hierarchy

The calibrated hierarchy is:

small in-sample residual $\Rightarrow$ descriptive compatibility,
 stability across admissible constructions $\Rightarrow$ construction-stable description,
 held-out evaluation under deployment information $\Rightarrow$ tested predictive generalization,
 validation across conditions or systems $\Rightarrow$ broader empirical regularity,
 discrimination among generators $\Rightarrow$ mechanistic support,
 interventional or causal design $\Rightarrow$ causal support.

The arrows indicate increasing requirements, not automatic implications. A target-containing coordinate can support an implicit constraint but cannot, by its residual alone, support independent prediction of the same contemporaneous target. One mesh cannot establish continuum validity; one trajectory cannot establish a system-wide invariant; and validation against a simulator alone is evidence about the simulated model, not automatically the physical system. Causal interpretation additionally requires a design or structural model capable of distinguishing intervention from association [32].

A concise reporting form is:

Within coordinate system C^* , constructed from the admissible record under provenance Π^* , relation R^* is supported for purpose q over domain Ω^* by evidence $\mathcal{E}^*$. The result does not resolve the alternatives or failure modes in $\mathcal{L}^*$.

S6 Additional controlled-study analyses

This note begins with analyses not shown in the main figures. The governing systems, discovery questions and primary outcomes are introduced only in Results; they are not repeated here.

S6.1 Derivative fidelity in the Lorenz study

The finite-difference study was designed to separate representational error from equation search. When derivatives are computed from sampled values, the library contains approximations to the coordinates required by the exact system. Coefficient differences therefore combine differentiation error, conditioning and sparse selection. Using the physical Δt is essential: differentiation with respect to sample index rescales the coefficients whenever the temporal resolution changes.

Supplying manufactured exact derivatives while leaving the search unchanged provides an upper-bound coordinate representation. The exact support remains the same and coefficient error decreases, confirming that the conventional Lorenz problem is contained in SIR. The example does not by itself establish an advantage over SINDy; it is a compatibility and provenance test.

S6.2 Partially observed Lorenz representation discovery

The representation-selection benchmark uses the same Lorenz dynamics as the generator-containment study but changes the information boundary and scientific task. The explanatory record contains only $x(t)$ and $y(t)$; $z(t)$ is retained as a supervised reconstruction target and is not admitted to the explanatory coordinate grammar.

A finite ontology was instantiated from local samples and admissible constructions of the two observed channels. Seventeen predeclared representation families were compared. Their union defines C_{all} , containing 38 evaluated coordinates. The representations were frozen as a finite benchmark before protected evaluation; the experiment therefore tests task-conditioned selection within a controlled ontology rather than open-ended ontology induction.

Forty-eight independently generated trajectories formed the development object. Twenty-four additional trajectories generated independently of the development cohort formed the protected confirmation object. Representation selection, sparse-regression settings and all task thresholds were fixed using the development cohort before the confirmation trajectories were evaluated.

PySINDy was retained as the common sparse relation estimator for every representation. For each candidate C , the estimator therefore solved the same downstream model-identification problem conditional on different coordinate information. This design separates representation selection from the numerical identity of the relation learner.

Under the accuracy contract,

$$q_{\text{acc}},$$

held-out reconstruction fidelity was the primary comparison criterion and the selected representation was the complete 38-coordinate construction,

$$C_{\text{acc}}^* = C_{\text{all}}.$$

Under the compactness contract,

$$q_{\text{comp}},$$

representations whose development performance lay within the declared practical fidelity floor were treated as task-equivalent before dimensionality was used to break the tie. The selected representation contained 12 coordinates,

$$|C_{\text{comp}}^*| = 12.$$

The complete representation and the selected compact representation were frozen before confirmation evaluation.

A conventional quadratic polynomial representation constructed from the same admitted raw observations was retained as a fixed baseline. A multilayer perceptron was also trained with identical architecture and optimization settings across the compared coordinate systems; this ablation tests whether utility of the selected coordinates is specific to sparse regression.

A robustness variant perturbed the admitted observations at the predeclared 1% benchmark noise level while retaining the same coordinate grammar. Consequently,

$$\mathcal{G}_{q,\text{clean}} = \mathcal{G}_{q,\text{noise}}$$

for this ablation even though the observational error object differs. This was intentional: the study measures robustness of the candidate representations already under consideration. It does not test the stronger adaptive-SIR procedure in which the changed error model would be allowed to revise derivative estimators, smoothing, denominator policies or other numerical coordinate realizations.

The exact coordinate inventory, benchmark random states, perturbation normalization, PySINDy settings and neural-network hyperparameters are retained in the frozen representation-benchmark manifest and should accompany the final code release. These are part of the coordinate and search provenance rather than free post hoc choices.

S6.3 Task-conditioned pendulum representations

The nonlinear-pendulum experiment isolates variation in the discovery contract while holding the observational object fixed. The common scientific object contains 32 trajectory realizations with state

$$(\theta(t), \omega(t)).$$

The restricted object-adapted ontology admits the measured state together with temporal-rate, trigonometric and quadratic-energy constructions. Both contracts therefore interrogate the same underlying trajectory ensemble rather than different simulated systems.

For the compression contract q_{comp} , the task utility rewards a low-dimensional shared description of each realized trajectory and permits one realization-specific scalar. The selected form is

$$\omega^2 = A(1 - \cos \theta) + B_i,$$

with a common coefficient A and realization-dependent intercept B_i . The recovered values satisfy

$$A = -3.6450, \quad B_i = 2E_i,$$

where E_i is the conserved energy of realization i . The near-perfect linear relation between fitted B_i and $2E_i$ provides the controlled interpretation of the realization-level scalar.

For the evolution contract q_{pred} , the selected representation must instead define an autonomous dynamical system using information available at propagation time. The recovered relation is

$$\dot{\theta} = \omega, \quad \dot{\omega} = -1.82245149 \sin \theta,$$

with the second coefficient compared against the known value -1.8225 .

Twenty-four of the 32 realizations were used during development. Structural selection and coefficient estimation were frozen before the remaining eight protected trajectories were evaluated. For those trajectories, only

$$(\theta(0), \omega(0))$$

was supplied to the frozen evolution law. The resulting system was propagated autonomously for 30 s, and pooled trajectory error was evaluated as

$$\text{RMSE}_{\theta}(T) = \left[\frac{1}{N_T} \sum_{k:t_k \leq T} (\theta_k - \hat{\theta}_k)^2 \right]^{1/2},$$

with the analogous definition for ω .

The two selected representations therefore differ because the contracts impose different scientific utilities and information requirements, not because the underlying observations differ. Under q_{comp} , the invariant is a qualified compression of the realized trajectories. Under q_{pred} , the invariant alone is insufficient because the contract requires autonomous temporal propagation.

S6.4 Stochastic explicit relation

The legacy SIR validation also considered

$$y(x) = h(x) + \alpha h'(x) + \zeta, \quad \alpha \sim U[1, 2],$$

with a small realization-specific additive term ζ . A library containing h, h' and h'' recovered the support dominated by h and h' , coefficient mass near unity for h , and the sampled distribution for α ; the fitted h'' coefficient was concentrated near zero. This study illustrates how coefficient distributions can be recovered for an explicit stochastic family. It is retained as extended validation because the oscillator example makes the same population-level point more cleanly in the main Article.

S6.5 Why one heat solution does not identify the heat equation

The separated field can be written as a product

$$u(x, t) = T(t)X(x), \quad T(t) = e^{-\alpha t}, \quad X(x) = \sin(m\pi x).$$

Consequently,

$$\frac{T'}{T} = -\alpha, \quad \frac{X''}{X} = -(m\pi)^2,$$

on their regular domains. The classical heat equation follows by equating the separation constants through $\alpha = (m\pi)^2\kappa$, but the temporal and spatial eigen-relations are individually complete descriptions of their respective factors. A parsimony score can therefore prefer either two-term relation.

The ambiguity can be reduced only when additional observations are coupled to a contract that requires the competing relations to generalize across the new variation. The multimode construction below provides an explicit example.

S6.6 Multimode observations resolve ambiguity under a shared-coefficient contract

The single-realization example in the preceding section establishes that residual fidelity alone cannot identify the heat equation from a separated solution. Here we ask what additional observational diversity is sufficient to discriminate among the surviving exact relations. We retain the same diffusivity κ , but observe a family of spatial modes

$$u_m(x, t) = \exp[-(m\pi)^2\kappa t] \sin(m\pi x), \quad m \in \mathcal{M},$$

where $\mathcal{M}$ contains two or more distinct positive integers. For each realization define

$$\alpha_m = (m\pi)^2\kappa.$$

Every individual field then satisfies the three exact relations

$$u_{m,t} + \alpha_m u_m = 0,$$

$$u_{m,xx} + (m\pi)^2 u_m = 0,$$

and

$$u_{m,t} - \kappa u_{m,xx} = 0.$$

Thus, considered realization by realization, the ambiguity identified in the main text remains: the temporal eigen-relation, spatial eigen-relation and coupled diffusion relation all have zero residual in exact arithmetic.

The distinction appears only when the discovery contract specifies which structure must be shared across the ensemble. Under a contract that permits realization-specific coefficients, the temporal family

$$u_{m,t} + a_m u_m = 0, \quad a_m = (m\pi)^2\kappa,$$

and the spatial family

$$u_{m,xx} + b_m u_m = 0, \quad b_m = (m\pi)^2,$$

remain valid shared-support representations. Their supports are common, but their coefficients vary systematically with the mode number m . Multiple realizations therefore do not, by themselves, eliminate these descriptions.

Now impose the stronger task requirement that a candidate relation have both common support and a common coefficient vector across all observed modes. Consider the candidate family

$$c_t u_t + c_0 u + c_x u_{xx} = 0.$$

For mode m , substitution of the exact field gives

$$[-c_t(m\pi)^2 \kappa + c_0 - c_x(m\pi)^2] u_m = 0.$$

For this relation to hold for two or more distinct values of m , the mode-independent and mode-dependent parts must vanish separately:

$$c_0 = 0, \quad c_t \kappa + c_x = 0.$$

Up to an arbitrary overall scaling, the surviving common-coefficient relation is therefore

$$u_t - \kappa u_{xx} = 0.$$

The multimode ensemble consequently changes the identifiability of the observational object under a shared-coefficient discovery contract. A single mode supports an equivalence class of exact differential descriptions, whereas variation in m eliminates the temporal and spatial eigen-relations as fixed-coefficient laws because their coefficients scale as m^2 . The diffusion coefficient κ , by contrast, is invariant across the realizations.

This example also illustrates why identifiability in SIR is task-conditioned rather than intrinsic to the numerical record. If the scientific objective is to describe each realization independently, the mode-specific eigen-relations remain legitimate and may even be more parsimonious. If the objective is instead to identify a relation transferable across a family of realizations with a common parameterization, the admissible model class changes and the coupled diffusion relation becomes distinguished. The additional observations are informative because the contract requires the representation to survive variation that breaks the competing alternatives.

More generally, observational degeneracy can be reduced by deliberately sampling directions along which candidate representations transform differently. Initial conditions, boundary conditions, spatial modes, parameter sweeps and controlled interventions can therefore be treated not only as additional data, but as experimental coordinates for discriminating among an existing qualified model set. In this sense, a retained ambiguity class in SIR identifies what kind of new observation would be useful for the next discovery cycle.

S7 DIII-D data construction, relational fitting and validation

S7.1 Canonical cohort and signal provenance

The canonical DIII-D analysis contains 62 discharge realizations. The fixed discharge list and source-file hashes are recorded in the accompanying provenance ledger. Each realization contains the eight quantities admitted to the structural search:

$$\{W_{\text{dia}}, P_{\text{NBI}}, n_e, I_p, \beta_N, q_{95}, \ell_i, \kappa\}.$$

These correspond to the source identifiers `pcdiamag3`, `pinj`, `density`, `ip`, `betan`, `q95`, `li` and `kappa`, respectively. Their observational roles are summarized in Table S4. The categories describe the available observational record and do not imply causal independence.

The mixture of measured, processed and equilibrium-reconstructed quantities is scientifically relevant. Shared upstream diagnostics or reconstruction procedures can induce dependence among nominally distinct coordinates. The selected network is consequently interpreted as an observational organization, not as a causal graph or proof that every selected coordinate contains independent physical information [33–36].

S7.2 Temporal alignment, reconstruction and standardization

For each discharge and signal, non-finite time–value pairs were removed, timestamps were sorted, and duplicate or non-increasing timestamps were discarded. The eight admitted signals were then restricted to their common temporal support. If $[t_{j,0}, t_{j,1}]$ denotes the retained interval for discharge j , the aligned grid was

$$t_{j,k} = t_{j,0} + k\Delta t_j, \quad \Delta t_j = \frac{t_{j,1} - t_{j,0}}{999}, \quad k = 0, \dots, 999.$$

Each realization therefore contains $N_j = 1000$ aligned samples.

The retained intervals span approximately 4.08–6.02 s, giving discharge-dependent physical spacings of approximately 4.08–6.03 ms. The canonical analysis therefore does not use a fixed 20 ms grid. Signals sampled more densely than the aligned grid were reduced using the provider’s local aggregation rule, whereas more sparsely sampled signals were interpolated onto the common support. The complete discharge-level intervals and spacings are recorded in the provenance ledger.

The provider represents time numerically in milliseconds. This convention rescales unstandardized temporal derivatives relative to derivatives formed in seconds. It does not change the reported standardized reconstruction error, because the target and retained derived coordinates are standardized before fitting. The time unit must nevertheless remain explicit in the frozen analysis package so that dimensional derivative products can be interpreted and reproduced correctly.

Primitive signals were standardized separately within each discharge:

$$\tilde{x}_r^{(j)}(t) = \frac{x_r^{(j)}(t) - \mu_r^{(j)}}{\sigma_r^{(j)}},$$

where $\mu_r^{(j)}$ and $\sigma_r^{(j)}$ are the temporal mean and standard deviation of quantity r in discharge j . The target and retained derived coordinates were likewise represented in standardized coordinates before regression. The fitted coefficients therefore describe a normalized observational representation rather than dimensional transport coefficients or universal physical constants.

Three trajectory representations were examined: aligned unsmoothed signals, univariate spline reconstructions and Rauch–Tung–Striabel reconstructions [17–19]. Derivatives and relational coordinates were constructed consistently from the corresponding trajectory representation. The exact reconstruction settings, derivative conventions and boundary policies must be tied to the frozen run manifest before the alternate reconstruction results are treated as final quantitative evidence.

S7.3 Candidate coordinates, selected support and coefficient fitting

The candidate library included retained observables, temporal derivatives, algebraic ratios and first-order trajectory-relational coordinates. Sparse structural selection targeted the standardized response

$$\tilde{y}^{(j)}(t) = \widetilde{\frac{dW_{\text{dia}}^{(j)}}{dt}}.$$

The canonical selected support contains seven nonconstant coordinates:

$$\psi = \left(D_\kappa W_{\text{dia}}, D_{\beta_N} W_{\text{dia}}, D_{\beta_N} \kappa, D_{\beta_N} \ell_i, \frac{q_{95}}{\kappa}, \dot{\beta}_N, \dot{\kappa} \right).$$

At the symbolic level,

$$D_g f = \frac{\dot{f}}{\dot{g}}$$

on the regular domain of the denominator. Any numerical shifting, masking or other denominator policy used in the implemented coordinate construction is part of the frozen coordinate provenance and must be reported with the final run.

The seven-coordinate support was shared across the discharge ensemble, but the coefficients were estimated separately for each realization:

$$\tilde{y}^{(j)}(t) = c_0^{(j)} + \sum_{r=1}^7 c_r^{(j)} \tilde{\psi}_r^{(j)}(t) + \varepsilon^{(j)}(t), \quad j = 1, \dots, 62. \quad (\text{S7.1})$$

The fitted object is therefore a shared relational support together with an empirical population of realization-specific coefficient vectors,

$$\mathbf{c}^{(j)} = (c_0^{(j)}, c_1^{(j)}, \dots, c_7^{(j)}).$$

This population may be represented by the empirical measure

$$\hat{\mathbb{P}}_{\mathbf{c}} = \frac{1}{62} \sum_{j=1}^{62} \delta_{\mathbf{c}^{(j)}}.$$

No parametric probability law is imposed on this collection.

The numerical values displayed in the relational-network figure are the cohort means of the discharge-specific coefficients. For the seven nonconstant terms, ordered as in Eq. (S7.1), these means are

$$(\bar{c}_1, \bar{c}_2, \bar{c}_3, \bar{c}_4, \bar{c}_5, \bar{c}_6, \bar{c}_7) = (0.248, 1.075, -1.070, -0.0150, -0.0106, 0.0509, 0.893),$$

with a negligible mean intercept. These values summarize the centre of the fitted coefficient population. They were not applied as one universal coefficient vector to reconstruct every discharge.

S7.4 Reconstruction metrics and coefficient summaries

For discharge j , let

$$e_{j,k} = \tilde{y}_{j,k} - \hat{\tilde{y}}_{j,k},$$

where $\hat{\tilde{y}}_{j,k}$ is reconstructed using the coefficient vector fitted for that discharge. The discharge-resolved root-mean-square error is

$$\text{RMSE}_j = \left[\frac{1}{N_j} \sum_{k=1}^{N_j} e_{j,k}^2 \right]^{1/2}.$$

The pooled mean-square and root-mean-square errors are

$$\text{MSE}_{\text{pooled}} = \frac{\sum_{j=1}^{62} \sum_{k=1}^{N_j} e_{j,k}^2}{\sum_{j=1}^{62} N_j}, \quad \text{RMSE}_{\text{pooled}} = \sqrt{\text{MSE}_{\text{pooled}}}.$$

All 62 discharges contain 1,000 retained samples, so the pooled metric contains 62,000 observations and assigns equal sample-count weight to each discharge.

For the canonical aligned representation,

$$\text{MSE}_{\text{pooled}} = 3.32 \times 10^{-3}, \quad \text{RMSE}_{\text{pooled}} = 5.76 \times 10^{-2}.$$

The main-text figure rounds the latter value to 0.058.

Applying the cohort-mean coefficient vector to the same seven-coordinate export instead gives a pooled RMSE of approximately 0.413. The reported 0.058 reconstruction error therefore quantifies discharge-specific coefficient fits on a shared support; it is not the error of one universal numerical equation.

The discharge-specific coefficient vectors should be supplied with the final frozen analysis as a machine-readable supplementary table, together with discharge-level reconstruction errors, coefficient summaries and the exact model and preprocessing identifiers used to generate them.

S7.5 Trajectory-reconstruction variants

The aligned, spline and Rauch–Tung–Striebel representations were introduced to test sensitivity to trajectory reconstruction rather than predictive generalization. Each variant requires its own derivative construction, coordinate evaluation, support record, realization-specific coefficients and error metrics.

Claims of support or coefficient stability across these variants should be based on frozen, like-for-like outputs. In particular, each reported error must identify whether it was obtained from discharge-specific coefficients or from a cohort summary vector, and whether the reported quantity is an MSE or an RMSE. Numerical spline and Rauch–Tung–Striebel results should remain provisional until their exact settings, coefficient semantics and run identifiers are tied to the frozen analysis manifest.

S7.6 Target containment and implicit rational structure

Two selected coordinates contain the contemporaneous response derivative:

$$D_{\kappa} W_{\text{dia}} \sim \frac{\dot{W}_{\text{dia}}}{\dot{\kappa}}, \quad D_{\beta_N} W_{\text{dia}} \sim \frac{\dot{W}_{\text{dia}}}{\dot{\beta}_N},$$

where the symbol $\sim$ indicates the corresponding symbolic quotient before any numerical denominator policy and standardization are applied. These coordinates are admissible for retrospective implicit description but cannot serve as independent predictors of the same contemporaneous target.

Substitution of the two target-containing coordinates into Eq. (S7.1) yields, for each discharge, an implicit relation of the schematic form

$$A_j(t) \dot{W}_{\text{dia}}^{(j)}(t) = B_j(t) + \varepsilon^{(j)}(t).$$

Here $A_j(t)$ and $B_j(t)$ depend on the realization-specific coefficients, normalization constants, non-target coordinates and the declared denominator policy. The selected model is therefore a population of low-order rational implicit closures sharing a common coordinate support, rather than a syntactic identity or an independently predictive explicit law.

A small retrospective residual establishes descriptive compatibility with this enriched coordinate family. It does not by itself establish that the relation contributes an independent constraint beyond the coordinate definitions, remains locally well conditioned, transfers to unseen discharges or represents a causal plasma mechanism.

S7.7 Coefficient conditioning and identifiability conditional on the selected support

The canonical DIII-D analysis assigns the same seven-coordinate support to all 62 discharges but estimates its coefficients separately for each realization. A low reconstruction error does not by itself establish that

these coefficients are uniquely or stably determined: correlated coordinates can permit substantial coefficient movement while producing similar fitted trajectories. We therefore evaluated coefficient identifiability conditional on the frozen support and design matrices archived in **D3D-SIR-62-ALIGNED-V1**. This analysis asks whether the individual coefficients and coefficient-space directions are numerically resolved within the selected representation. It does not test whether the support itself is uniquely selected, whether it transfers to unseen discharges or whether its coefficients have a mechanistic interpretation.

For discharge j , let

$$\mathbf{X}^{(j)} = \begin{bmatrix} \tilde{\psi}_1^{(j)} & \dots & \tilde{\psi}_7^{(j)} \end{bmatrix} \in \mathbb{R}^{1000 \times 7}$$

be the frozen standardized design matrix and let $\mathbf{y}^{(j)} \in \mathbb{R}^{1000}$ be the standardized $\dot{W}_{\text{dia}}$ target. In manuscript display order, the seven columns are

$$\left(D_{\kappa} W_{\text{dia}}, D_{\beta_N} W_{\text{dia}}, D_{\beta_N} \kappa, D_{\beta_N} \ell_i, \frac{q_{95}}{\kappa}, \dot{\beta}_N, \dot{\kappa} \right).$$

The fitted model is

$$\mathbf{y}^{(j)} = c_0^{(j)} \mathbf{1} + \mathbf{X}^{(j)} \mathbf{c}^{(j)} + \boldsymbol{\varepsilon}^{(j)}, \quad j = 1, \dots, 62, \quad (\text{S7.2})$$

where $\mathbf{c}^{(j)} = (c_1^{(j)}, \dots, c_7^{(j)})^\top$. The intercept was retained in every coefficient fit but excluded from the singular-value analysis of dependence among the seven nonconstant coordinates. Source-matrix and manuscript display orders were tracked separately throughout the audit, and coefficients and covariance matrices were permuted only through the recorded mapping.

Least-squares refitting reproduced the archived discharge-specific coefficients to a maximum absolute discrepancy of 2.22×10^{-16} . The resulting pooled reconstruction RMSE was

$$\text{RMSE}_{\text{pooled}} = 0.0575846725,$$

identical to the canonical value at the reported precision. Applying the cohort-mean coefficient vector to every discharge instead gave

$$\text{RMSE}_{\text{mean}} = 0.4125238316.$$

The conditioning analysis therefore concerns the realization-specific coefficient family that produces the lower reconstruction error, rather than the cohort-mean vector.

S7.7.1 Rank, correlation and singular geometry

For each discharge, we computed

$$\mathbf{X}^{(j)} = \mathbf{U}^{(j)} \boldsymbol{\Sigma}^{(j)} \mathbf{V}^{(j)\top},$$

where

$$\boldsymbol{\Sigma}^{(j)} = \text{diag} \left(\sigma_1^{(j)}, \dots, \sigma_7^{(j)} \right), \quad \sigma_1^{(j)} \geq \dots \geq \sigma_7^{(j)}.$$

All 62 matrices retained numerical rank seven at relative tolerances 10^{-12} , 10^{-10} , 10^{-8} and 10^{-6} . The spectral condition number

$$\kappa_2(\mathbf{X}^{(j)}) = \frac{\sigma_1^{(j)}}{\sigma_7^{(j)}}$$

had median 41.32 and maximum 71.32. The median normalized terminal singular value was

$$\text{median}_j \left(\frac{\sigma_7^{(j)}}{\sigma_1^{(j)}} \right) = 0.0242,$$

and its minimum across the cohort was 0.0140. The fitted matrices are therefore full rank and moderately, rather than catastrophically, ill-conditioned.

Full rank does not imply that all coordinate directions are comparably resolved. The strongest pairwise dependence occurred between $D_{\beta_N} W_{\text{dia}}$ and $\dot{\beta}_N$, whose absolute correlation had cohort median 0.828, attained

a maximum of 0.952, and exceeded 0.9 in approximately 31% of the discharges. The weakest right-singular direction was most often dominated by $D_{\beta_N} \kappa$, which had the largest absolute loading in approximately 77% of the discharges. These results identify a recurrent compensating direction in the fitted coordinate space, but do not make that direction a physical relation.

S7.7.2 Sensitivity to weak singular directions

The initial conditioning analysis evaluated truncated singular-value solutions only at relative thresholds no larger than 10^{-6} . Because every discharge satisfied

$$\frac{\sigma_7^{(j)}}{\sigma_1^{(j)}} \geq 0.0140,$$

those thresholds removed no singular direction. The resulting near-machine-precision coefficient changes represented recomputation of the same full-rank solution rather than a meaningful perturbation test.

We therefore repeated the analysis using relative singular-value thresholds

$$10^{-3}, \quad 5 \times 10^{-3}, \quad 10^{-2}, \quad 2 \times 10^{-2}, \quad 5 \times 10^{-2}, \quad 10^{-1},$$

and by explicitly retaining ranks seven, six, five and four. For a retained rank r , the truncated solution was formed from the first r singular directions of the centered design matrix, with the intercept treated consistently with the canonical least-squares fit.

Rank-seven reconstruction reproduced the canonical coefficients to numerical precision. Removing only the weakest singular direction, however, produced

$$\text{median}_j \frac{\left\| \hat{\mathbf{c}}_{(6)}^{(j)} - \hat{\mathbf{c}}_{(7)}^{(j)} \right\|_2}{\left\| \hat{\mathbf{c}}_{(7)}^{(j)} \right\|_2} = 0.799,$$

while the median absolute change in discharge-level RMSE was

$$\text{median}_j \left| \text{RMSE}_{(6)}^{(j)} - \text{RMSE}_{(7)}^{(j)} \right| = 0.0310.$$

Of the 62 discharges, 18 exhibited large coefficient movement while retaining a comparatively stable reconstruction under the declared criterion, 40 exhibited large coefficient movement together with a measurable reconstruction change, and four retained both stable coefficients and stable reconstruction. Retaining only five singular directions increased the median relative coefficient change to 0.828 and the median absolute RMSE change to 0.0413; retaining four increased them to 0.886 and 0.189, respectively.

A complementary ridge path,

$$\hat{\mathbf{c}}_\lambda^{(j)} = \arg \min_{\mathbf{c}} \left\{ \left\| \mathbf{y}^{(j)} - c_0^{(j)} \mathbf{1} - \mathbf{X}^{(j)} \mathbf{c} \right\|_2^2 + \lambda \|\mathbf{c}\|_2^2 \right\},$$

with the intercept unpenalized and $\lambda = \lambda_{\text{rel}}(\sigma_1^{(j)})^2$, gave the same qualitative result. The median relative coefficient change increased from 0.011 at $\lambda_{\text{rel}} = 10^{-5}$ to 0.102 at 10^{-4} , 0.480 at 10^{-3} and 0.779 at 10^{-2} . Thus, the selected linear span supports stable reconstruction, but the individual coefficient decomposition remains sensitive along its weakest directions.

S7.7.3 Within-discharge uncertainty and coefficient heterogeneity

Because the 1,000 observations within each discharge are temporally ordered, coefficient uncertainty was estimated using a circular moving-block bootstrap rather than independent-row resampling. The primary

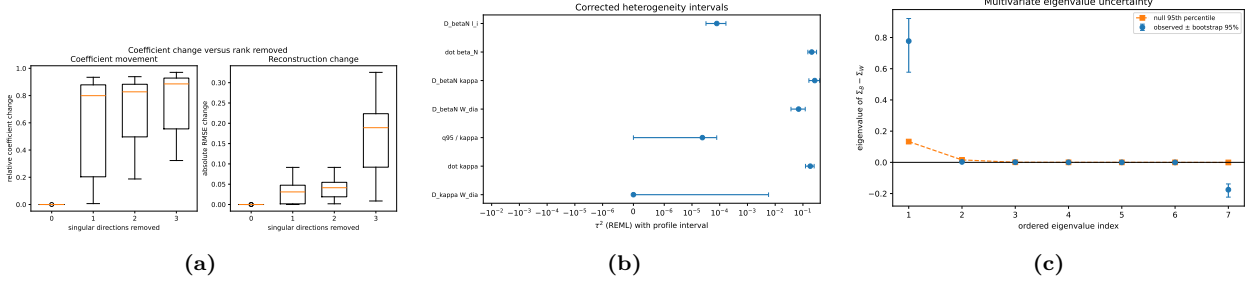


Figure S1: Conditional identifiability of the DIII-D coefficient family. **a**, Relative coefficient-vector change and absolute reconstruction RMSE change after removal of the weakest singular directions. Removing one direction produces substantial coefficient movement despite a more moderate change in reconstruction error. **b**, Restricted-maximum-likelihood estimates of between-discharge variance τ^2 , with profile-likelihood intervals. Five coefficients retain positive between-discharge variation under block-length and leave-one-discharge-out sensitivity tests; two are dominated by within-discharge uncertainty. **c**, Ordered eigenvalues of $\Sigma_B - \Sigma_W$. Intervals are obtained from 5,000 discharge-bootstrap replicates, and null thresholds from 5,000 matched simulations with no between-discharge coefficient variation. Two coefficient-space directions satisfy both the bootstrap and null criteria for resolution.

analysis used $B = 1000$ bootstrap replicates per discharge. Primary block lengths were chosen from the largest resolved correlation scale among the seven features and the canonical residual, with median block length 58 samples and range 13–116 samples across the cohort. Short- and long-block sensitivity analyses used $B = 300$ replicates at approximately one-half and twice the primary block length.

For coefficient r in discharge j , let $\hat{s}_{W,jr}^2$ denote its primary block-bootstrap variance. The observed between-discharge variance was

$$\hat{s}_{B,r}^2 = \frac{1}{61} \sum_{j=1}^{62} \left(\hat{c}_r^{(j)} - \bar{c}_r \right)^2,$$

and the mean within-discharge variance was

$$\bar{s}_{W,r}^2 = \frac{1}{62} \sum_{j=1}^{62} \hat{s}_{W,jr}^2.$$

For descriptive comparison, we report

$$H_r = \frac{\hat{s}_{B,r}^2}{\bar{s}_{W,r}^2}, \quad (\text{S7.3})$$

but coefficient classification was not based on this ratio alone.

Coefficient-wise between-discharge variation was estimated using a heteroscedastic random-effects model,

$$\hat{c}_r^{(j)} = \mu_r + u_r^{(j)} + e_r^{(j)},$$

where

$$u_r^{(j)} \sim \mathcal{N}(0, \tau_r^2), \quad e_r^{(j)} \sim \mathcal{N}(0, \hat{s}_{W,jr}^2).$$

The primary estimate of τ_r^2 used restricted maximum likelihood. With

$$v_{jr} = \hat{s}_{W,jr}^2 + \tau_r^2, \quad w_{jr} = v_{jr}^{-1}, \quad \hat{\mu}_r = \frac{\sum_j w_{jr} \hat{c}_r^{(j)}}{\sum_j w_{jr}},$$

the profiled restricted log-likelihood was

$$\ell_{R,r}(\tau_r^2) = -\frac{1}{2} \left[\sum_{j=1}^{62} \log v_{jr} + \log \left(\sum_{j=1}^{62} w_{jr} \right) + \sum_{j=1}^{62} w_{jr} \left(\hat{c}_r^{(j)} - \hat{\mu}_r \right)^2 \right]. \quad (\text{S7.4})$$

The earlier audit implementation omitted the $\log(\sum_j w_{jr})$ term and was therefore profile maximum likelihood, not REML. The corrected analysis uses Eq. (S7.4) as the primary estimator and retains profile maximum likelihood only as a sensitivity calculation.

A coefficient was classified as robustly resolved only when its estimated between-discharge variance remained positive under the primary analysis, its profile-likelihood interval excluded zero, the conclusion persisted under short- and long-block variance estimates, and no individual discharge changed the classification under leave-one-discharge-out refitting. Five of the seven coefficients satisfied these requirements:

$$\dot{\kappa}, \quad D_{\beta_N} W_{\text{dia}}, \quad D_{\beta_N} \kappa, \quad \dot{\beta}_N, \quad D_{\beta_N} \ell_i.$$

For $D_{\kappa} W_{\text{dia}}$ and q_{95}/κ , within-discharge uncertainty dominated the observed cross-discharge dispersion.

For each of the five robustly resolved coefficients, the estimated τ_r^2 remained positive in every leave-one-discharge-out fit, and no single realization determined the classification. The two uncertainty-dominated terms behaved differently. For $D_{\kappa} W_{\text{dia}}$, the primary REML estimate was zero and the leave-one-out estimate remained at the boundary throughout. For q_{95}/κ , the point estimate was positive but small, its profile interval included zero and the mean within-discharge variance exceeded the observed between-discharge variance. A positive point estimate was therefore not treated as resolved heterogeneity.

S7.7.4 Multivariate coefficient-space identifiability

Coefficient-wise analysis can miss stable linear combinations when individual coefficients compensate for one another. Let

$$\Sigma_B = \text{Cov}_j [\hat{\mathbf{c}}^{(j)}]$$

be the observed covariance of the 62 coefficient vectors, and let

$$\bar{\Sigma}_W = \frac{1}{62} \sum_{j=1}^{62} \hat{\Sigma}_W^{(j)}$$

be the mean within-discharge bootstrap covariance. We examined the symmetrized difference

$$\Sigma_R = \Sigma_B - \bar{\Sigma}_W, \tag{S7.5}$$

whose positive eigenvalues are point estimates of coefficient-space directions with cross-discharge variance exceeding average fitting uncertainty.

The point estimate of Σ_R contained five positive eigenvalues. Positivity alone is insufficient because sampling variation can generate positive eigenvalues even in the absence of genuine between-discharge variation. We therefore performed 5,000 discharge-level bootstrap replicates, carrying the corresponding within-discharge covariance matrix with each resampled discharge. We also generated 5,000 matched null replicates in which all discharges shared a common latent coefficient vector and differed only through resampled within-discharge bootstrap deviations.

An ordered eigenvalue was called robustly resolved only when its observed value was positive, its discharge-bootstrap lower 95% bound was positive and it exceeded the 95th percentile of the corresponding null distribution. Two of the five positive point estimates met all three requirements.

The leading resolved direction was stable under discharge resampling. For the two-dimensional resolved subspace, the median bootstrap principal angles were approximately 1.6° and 20.2° , while the 95th percentile of the maximum angle was 38.1° . The leading direction is therefore more stably oriented than the secondary direction. These directions are statistically resolved combinations of coefficients in the selected representation; they are not interpreted as latent plasma variables or physical laws.

1799 S7.7.5 Qualified interpretation

1800 The corrected audit supports mixed coefficient identifiability. All 62 seven-column design matrices are full
 1801 rank and moderately conditioned, and the realization-specific fits reproduce the standardized target with low
 1802 retrospective error. Five individual coefficients exhibit cross-discharge variation that remains resolved after
 1803 temporal block resampling, REML estimation, block-length sensitivity and leave-one-discharge-out analysis.
 1804 Two coefficients do not vary across discharges by more than their estimated within-discharge uncertainty.
 1805 At the multivariate level, two coefficient-space directions remain resolved after discharge-bootstrap and
 1806 no-heterogeneity null testing.

1807 At the same time, removing the weakest singular direction changes the fitted coefficient vector substantially
 1808 while producing a more moderate change in reconstruction error. The common seven-coordinate architecture
 1809 can therefore be more stable than its complete coefficient decomposition. The justified conclusion is not that
 1810 all seven coefficients constitute a uniformly identified stochastic family, but that the selected support contains
 1811 a mixture of robustly varying and uncertainty-dominated coefficient directions.

1812 This analysis is conditional on the selected support. It does not establish that the seven-coordinate structure
 1813 is uniquely preferred over alternative supports, that any coefficient is a dimensional physical constant, that
 1814 the relation transfers predictively to unseen discharges or that the observed coefficient variation has a causal
 1815 or mechanistic plasma interpretation. Conditioning of the target-containing implicit relation, including its
 1816 primitive quotient denominators and the coefficient multiplying $\dot{W}_{\text{dia}}$ after algebraic elimination, is evaluated
 1817 separately.

1818 S7.8 Target-free reconstruction contract and structural transfer

1819 The second DIII-D study uses the same 62-discharge scientific object but changes the discovery contract before
 1820 coordinate construction. The reconstruction target is I_p . Accordingly, the information operator excludes I_p
 1821 and every coordinate whose construction depends on I_p from the explanatory representation:

$$I_p \notin C_{\text{rec}}, \quad \text{Desc}(I_p) \cap C_{\text{rec}} = \emptyset.$$

1822 Here $\text{Desc}(I_p)$ denotes target-dependent descendants generated by the coordinate grammar. This exclusion is
 1823 applied before structural search rather than checked after model fitting.

1824 The restricted reconstruction ontology contains target-free primitive observables and admitted algebraic,
 1825 differential and trajectory-relational constructions. After numerical admissibility and duplicate handling, the
 1826 evaluated relational frontier contains 141 coordinates. We denote the full representation by

$$C_{\text{REL141}}.$$

1827 Seven discharges constitute the structural-development cohort. Sparse structural selection is performed only
 1828 on this cohort and selects a ten-coordinate support,

$$C_{\text{REL10}} \subset C_{\text{REL141}}, \quad |C_{\text{REL10}}| = 10.$$

1829 The support is then frozen. No discharge in the remaining 55-member external cohort contributes to the
 1830 choice of retained coordinates.

1831 Two comparison representations are used. REL141 retains all 141 admissible relational coordinates and
 1832 measures the reconstruction performance available from the dense relational basis. RAW10 is a ten-coordinate
 1833 representation constructed at matched nominal dimensionality without the selected relational expansion.
 1834 These comparisons separate coordinate compression from the more ordinary question of whether I_p is
 1835 reconstructible at all.

1836 For an external discharge j , let the trajectory contain N_j ordered samples and define the calibration index

$$N_j^{\text{cal}} = \lfloor 0.8N_j \rfloor.$$

The transferred support is held fixed while coefficients are estimated from

$$\{t_{j,k} : 1 \leq k \leq N_j^{\text{cal}}\}.$$

Reconstruction is evaluated on

$$\{t_{j,k} : N_j^{\text{cal}} < k \leq N_j\}.$$

Thus the experiment transfers coordinate structure but recalibrates realization-specific coefficients.

For representation C , discharge-level reconstruction error is

$$\text{RMSE}_j(C) = \left[\frac{1}{N_j^{\text{eval}}} \sum_{k \in \mathcal{E}_j} \left(\tilde{I}_{p,j,k} - \widehat{\tilde{I}}_{p,j,k}(C) \right)^2 \right]^{1/2},$$

where $\mathcal{E}_j$ denotes the protected final 20% and $N_j^{\text{eval}} = |\mathcal{E}_j|$. The pooled RMSE is computed from all protected residuals across the 55 external discharges.

The reported pooled values are

$$\text{RMSE}_{\text{REL141}} = 0.114, \quad \text{RMSE}_{\text{REL10}} = 0.126, \quad \text{RMSE}_{\text{RAW10}} = 0.182.$$

REL10 has lower discharge-level reconstruction error than RAW10 for

$$46/55$$

external discharges. The nominal coordinate reduction from 141 to 10 is 92.9%, while the effective degrees of freedom decrease from approximately 99 for the dense relational representation to 10.6 for REL10.

These values should not be interpreted as zero-shot prediction. The coordinate support transfers across discharges, but coefficients are recalibrated from the calibration portion of each external trajectory. Likewise, the final 20% is a reconstruction holdout rather than a future-forecast test under a strictly causal preprocessing boundary. The qualified claim is therefore structural: a compact target-free relational representation selected on seven discharges retains reconstruction utility across a substantially larger external cohort.

The exact development-discharge identifiers, complete 141-coordinate inventory, selected REL10 support, RAW10 definition and frozen search settings should be reported in the accompanying provenance table and machine-readable run manifest.

S8 Provenance, ontology refinement and prospective search architectures

The results in this Article use finite, frozen relational ontologies, but the general SIR construction permits the ontology and search policy themselves to be refined during development. Such refinement is admissible only when the branch history and evaluation boundary remain explicit.

S8.1 Provenance-preserving ontology refinement

Let

$$\mathcal{G}_q^{(0)} \longrightarrow \mathcal{G}_q^{(1)} \longrightarrow \dots \longrightarrow \mathcal{G}_q^{(m)}$$

denote a sequence of ontology revisions. A revision can be motivated by mathematical singularity, numerical instability, redundant constructions, uncertainty amplification, newly recognized scientific structure or the promotion of a useful composite quantity to a reusable coordinate.

Each revision defines a new discovery branch. If

$$\mathcal{G}_q^{(r)} \neq \mathcal{G}_q^{(r+1)},$$

the provenance record must retain the parent ontology, the evidence that motivated the revision, added or removed constructors, changed admissibility rules and any corresponding change in Σ_q or $\mathcal{B}_q$. Protected evaluation evidence cannot be used to construct $\mathcal{G}_q^{(r+1)}$ while remaining protected for that branch.

The same principle applies to numerical realization. Changing a smoothing scale, derivative estimator, quotient regularization, mask or normalization can change the evaluated coordinate even when its semantic label is unchanged. A valid revision therefore repeats

object realization $\rightarrow$ coordinate construction $\rightarrow$ relation search $\rightarrow$ qualification

rather than modifying only the final fitted relation.

S8.2 Corpus-informed and learned proposal priority

An admissible scientific corpus can guide search without determining the scientific verdict. Let

$$\mathcal{H}_q$$

contain task-admissible external knowledge such as peer-reviewed publications, curated databases, established theoretical relations or previous qualified discoveries. A proposal policy can use this information to construct

$$\pi_q(\xi \mid X_q, \mathcal{G}_q, \mathcal{H}_q),$$

where ξ is an admissible partial or complete construction.

Priority can reflect semantic proximity, known physical structure, dimensional or symmetry arguments, numerical conditioning, computational cost, historical success in related systems or a learned estimate of expected utility. It can be produced by conventional heuristics, probabilistic search, scientific foundation models or tool-using agents.

The evidential distinction is

$$\boxed{\text{proposal priority} \neq \text{validation evidence}}.$$

A construction that is common in the literature can justifiably be searched earlier without receiving additional evidential weight in $\mathcal{V}_q$. Conversely, low-priority admissible constructions should remain accessible when novelty is part of the discovery objective.

S8.3 Qualified recursion

A qualified output can itself become part of a subsequent scientific object, provided that its epistemic status is inherited rather than erased. If

$$\mathcal{Q}_{q_i}^* = (\mathcal{K}_{q_i}, C_i^*, R_i^*, \Omega_i^*, \Pi_i^*, \mathcal{A}_i^*, \mathcal{E}_{\text{val},i}^*, \mathcal{L}_i^*),$$

then a higher-order object may be formed schematically as

$$X^{(2)} = \text{Obj}^{(2)}(\mathcal{Q}_{q_1}^*, \dots, \mathcal{Q}_{q_m}^*).$$

The new object inherits the supported domains, uncertainty, information availability, unresolved ambiguity and provenance of its components.

Such recursion can operate on relations, coefficient populations, residual fields, regime labels, equivalence classes, conditioning maps or validation results. An ambiguity class should remain an ambiguity class unless new observations distinguish its members; an ill-conditioned relation should not be silently promoted into a reliable predictor; and a target-containing descriptive coordinate remains target-containing when passed to a higher-order analysis.

The resulting architecture permits increasingly integrated scientific representations while keeping the strength of every derived claim bounded by the evidence inherited from its constituents.

S8.4 Agent-assisted discovery and audit

Scientific agents can participate in ontology construction and search by proposing mathematical interpretations of a record, retrieving relevant scientific precedent, constructing candidate coordinates, detecting admissibility violations, proposing numerical realizations, generating null models or identifying observations capable of separating surviving ambiguity classes.

An agentic search remains compatible with SIR only if its actions are included in provenance. At minimum, the audit state should retain the scientific question, information boundary, ontology version, external knowledge sources, proposed and rejected constructions, rejection reasons, search priorities, development and protected partitions, evaluated frontier and final qualification evidence.

This requirement separates open-ended hypothesis generation from open-ended claim generation. An agent can substantially enlarge or reorder the space of scientific hypotheses while the contract and validation protocol continue to determine which claims the observational system actually supports.

Table S1: Illustrative operational objective classes. These examples are not intended as an exhaustive taxonomy. Rather, they show how familiar mathematical representations acquire utility under different information and validation contracts. Movement from description toward decision generally introduces additional requirements; it does not imply that the weaker objectives are scientifically inferior or that every useful representation should support all four.

Illustrative objective	Operational role	Canonical examples and practical utility
Description / compression	Represent the realized scientific object by a simpler coordinate system, constraint, reduced manifold or parameter family while preserving distinctions material to the question. Contemporaneous and retrospectively constructed quantities can be admissible.	<i>Equation of state:</i> $PV - nRT = 0$ compresses thermodynamic equilibrium states into a relational constraint without privileging any variable as the response [37]. <i>Principal-component representation:</i> a high-dimensional set of observations is represented by a lower-dimensional subspace preserving dominant variation [38].
Reconstruction / compression	Recover a withheld, latent or unavailable component of the contemporaneous scientific object from information that remains admissible, preferably through a representation substantially simpler than the original observational description.	<i>Computed tomography:</i> an internal spatial field is reconstructed from measured line integrals [39]. <i>State estimation:</i> a latent dynamical state is reconstructed from noisy and partial observations, as in Kalman filtering [?].
Prediction	Estimate a quantity outside the present information set, usually at a future time, parameter value or condition. Every coordinate used by the model must be constructible without access to the protected information.	<i>Numerical weather prediction:</i> an estimated atmospheric state and dynamical representation are propagated forward to forecast future conditions [41]. <i>Time-series forecasting:</i> structured dependence on an observed history is used to estimate future observations [40].
Decision / control	Select an action whose application changes the subsequent system trajectory. Utility is therefore attached to realized decisions, costs, constraints or closed-loop behavior rather than to reconstruction or forecast error alone.	<i>Linear-quadratic control:</i> feedback actions are selected to minimize an accumulated state-control cost [42]. <i>Model-predictive control:</i> a constrained finite-horizon optimization is repeatedly solved and the first action is applied in closed loop [43].

Point	t_i	$\gamma(t_i)$	$(p(t_i), p(t_i + \pi))$	$(p_s(t_i), p_s(t_i + \pi))$	Tangent angle
A	0.5782	(0.7383, 1.8538)	(7.982, -1.271)	(0.367, 0.391)	45.3°
B	1.4994	(-1.6403, 0.6768)	(0.756, -0.661)	(1.083, 3.155)	70.6°
C	2.6590	(0.4337, -2.0455)	(-6.844, -1.502)	(1.625, 2.844)	25.3°

Table S3: Crossing-level diagnostics for the controlled multi-crossing trajectory. The second preimage associated with each reported t_i is $t_i + \pi$. The tangent angle is the acute angle between the two unoriented tangent lines in the planar observational projection. Both the classical and shifted relational coordinates distinguish the paired passages at all three crossings.

Table S4: DIII-D quantities admitted to the canonical structural search. Native archive units must be taken from the frozen source metadata rather than inferred from the symbols or source names. The fitted coefficients refer to discharge-wise standardized coordinates and are therefore not dimensional physical constants.

Symbol	Source key	Observational role	Native units	
W_{dia}	pcdiamag3	Diamagnetic stored-energy proxy and target-bearing quantity	Frozen metadata	archive
P_{NBI}	pinj	Neutral-beam actuation	Frozen metadata	archive
n_e	density	Global electron-density state	Frozen metadata	archive
I_p	ip	Plasma-current state	Frozen metadata	archive
β_N	betan	Normalized pressure state	Dimensionless	
q_{95}	q95	Edge safety factor from equilibrium reconstruction	Dimensionless	
ℓ_i	li	Internal-inductance and current-profile descriptor	Dimensionless	
κ	kappa	Plasma elongation from equilibrium reconstruction	Dimensionless	

Table S5: Canonical seven-coordinate support and cohort-mean coefficients. Coefficients are means of the realization-specific fits in standardized coordinates and are shown for descriptive orientation only.

Coordinate	Cohort mean
$D_\kappa W_{\text{dia}}$	0.248
$D_{\beta_N} W_{\text{dia}}$	1.075
$D_{\beta_N} \kappa$	−1.070
$D_{\beta_N} \ell_i$	−0.0150
q_{95}/κ	−0.0106
$\dot{\beta}_N$	0.0509
$\dot{\kappa}$	0.893

Table S6: Corrected coefficient-wise identifiability results. H is the ratio of observed between-discharge variance to mean within-discharge block-bootstrap variance. The reported τ^2 values and intervals are from the heteroscedastic REML analysis. A coefficient was classified as robustly resolved only after profile-interval, block-length and leave-one-discharge-out confirmation.

Coordinate	H	$\hat{\tau}^2$	95% profile interval	Classification
$D_\kappa W_{\text{dia}}$	0.184	0	$[0, 5.77 \times 10^{-3}]$	Within-discharge uncertainty dominates
$\dot{\kappa}$	10.1	0.184	$[0.124, 0.254]$	Robustly resolved
q_{95}/κ	0.446	2.37×10^{-5}	$[0, 7.80 \times 10^{-5}]$	Within-discharge uncertainty dominates
$D_{\beta_N} W_{\text{dia}}$	1.20	0.0697	$[0.0371, 0.121]$	Robustly resolved
$D_{\beta_N} \kappa$	2.64	0.267	$[0.164, 0.400]$	Robustly resolved
$\dot{\beta}_N$	11.4	0.208	$[0.151, 0.310]$	Robustly resolved
$D_{\beta_N} \ell_i$	3.75	7.84×10^{-5}	$[3.20 \times 10^{-5}, 1.67 \times 10^{-4}]$	Robustly resolved

Table S7: Uncertainty analysis of multivariate coefficient-space directions. Eigenvalues are ordered from largest to smallest. Bootstrap intervals are based on 5,000 discharge-level replicates; the null threshold is the corresponding 95th percentile under no between-discharge coefficient variation.

Index	Observed	Bootstrap 95% interval	Null 95%	Classification
1	7.77×10^{-1}	$[5.777 \times 10^{-1}, 9.218 \times 10^{-1}]$	1.331×10^{-1}	Robustly resolved
2	2.895×10^{-3}	$[1.304 \times 10^{-3}, 4.780 \times 10^{-3}]$	1.567×10^{-2}	Positive point estimate only
3	8.894×10^{-4}	$[3.604 \times 10^{-4}, 1.452 \times 10^{-3}]$	5.816×10^{-4}	Robustly resolved
4	1.491×10^{-4}	$[-4.312 \times 10^{-6}, 4.096 \times 10^{-4}]$	1.113×10^{-4}	Positive point estimate only
5	5.097×10^{-5}	$[-3.179 \times 10^{-4}, 7.832 \times 10^{-5}]$	-1.326×10^{-5}	Positive point estimate only
6	-4.341×10^{-4}	$[-7.402 \times 10^{-4}, -2.819 \times 10^{-4}]$	-1.500×10^{-4}	Not resolved
7	-1.753×10^{-1}	$[-2.229 \times 10^{-1}, -1.383 \times 10^{-1}]$	-6.509×10^{-4}	Not resolved

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